

Jindo: Practical Lattice-Based Polynomial Commitments for Client-Side Proving

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Abstract—We present **Jindo**, a new lattice-based polynomial commitment scheme (PCS) optimized for client-side proving, which requires fast prover performance while supporting evaluation hiding and flexible parameter regimes. To achieve this, we build **Jindo** on the previous lattice-based PCSs **CELPC** (CRYPTO’24) and **Greyhound** (CRYPTO’24), which exhibit fast proving performance but have several limitations for client-side proving. We resolve these limitations by developing a new polynomial evaluation protocol that supports multilinear polynomials, diverse field choices, sublinear masking overhead for evaluation hiding, and cube-root verification and communication complexity.

Our implementation shows that **Jindo** improves upon **CELPC** by about an order of magnitude across all metrics, including proof generation, verification, and proof size. Compared with the recent evaluation hiding code-based PCS **PIP-FRI** (USENIX’26), **Jindo** provides an order of magnitude faster proof generation while yielding similar verification and communication costs. Furthermore, when compiling the PIOP of **Buckler** (CCS’25) to prove the validity of an RLWE sample, **Jindo** outperforms **CELPC** by an order of magnitude across all metrics. Compared with **Lazer** (CCS’24) for proving the same lattice relation, **Jindo** achieves an order of magnitude faster proof generation and two orders of magnitude faster verification.

Keywords—polynomial commitment, zero-knowledge argument, lattice

1. Introduction

A zero-knowledge argument of knowledge (ZKAoK) is a cryptographic protocol that allows a prover to convince a verifier that it knows a secret witness for a given statement without revealing any information beyond the validity of the statement. It has become a core building block for various privacy-preserving protocols where selective disclosure of a client input to a server is required for authentication while preserving client privacy. Such examples include e-voting [ABGS23], anonymous credentials [BLNS23], secure aggregation [BGL⁺23], and privacy-preserving signatures [AGJ⁺24]. More recently, there has been growing demand for ZKAoKs in real-world applications such as identity verification [Eur26, Nat26], digital

wallets [Eur24, KD25], and proof of humanity for AI security [Wor24]. In these protocols, the computational overhead for the client is primarily dominated by proof generation. Therefore, constructing efficient ZKAoK systems optimized for client-side proving scenarios is required to practically deploy these applications.

In recent years, the paradigm for ZKAoK construction has rapidly evolved, driven by the polynomial interactive oracle proof (PIOP) framework [BFS20, CHM⁺20], which provides a systematic and unified approach to designing ZKAoKs. In this framework, a proof system is first designed as an interactive oracle proof (IOP) [BSCS16], where in each round the prover sends oracles representing polynomials and the verifier responds with random challenges or evaluation queries to the polynomial oracles. Such an IOP is then compiled into a ZKAoK using a polynomial commitment scheme (PCS), a variant of a commitment scheme that enables efficient proof of evaluation of committed polynomials over finite fields. The key advantage of PIOP-based constructions is their broad support for various statements, ranging from general arithmetic circuits based on R1CS representations [CHM⁺20] to lookup arguments [AST24] suitable for instruction set architectures. For a proof system constructed via PIOP compilation, the efficiency of the PCS used in the compilation determines the overall performance, including proof generation, verification, and proof size. Therefore, the PIOP framework reduces the construction of efficient ZKAoK systems to the construction of an efficient PCS, making PCS design a major research focus for improving the overall performance of PIOP-based ZKAoKs.

To construct efficient PCSs, various designs have been proposed based on different cryptographic assumptions. PCSs were initially developed using pairing-based assumptions [KZG10, Lee21] or discrete logarithm-based assumptions [BBB⁺18, WTS⁺18], leveraging the algebraic structure of elliptic curves. Since these constructions do not provide post-quantum security, code-based constructions [GLS⁺23, ZXZS20, GLH⁺25, ACFY25] have been widely studied as an alternative, offering post-quantum security based on symmetric primitives. More recently, there has been growing interest in lattice-based constructions [CMNW24, NS24, HSS24, KLNO24, KLNO25], as they also offer post-quantum security. However, existing PCS designs primarily target blockchain applications such

as zkVMs and zkRollups [Eth, Pol], which prioritize small proof sizes and fast verification to address scalability issues in blockchain. Moreover, this design trend often overlooks proof generation speed and zero-knowledgeness, both of which are key requirements for client-side proving scenarios. Therefore, designing PCSs tailored to client-side proving scenarios still requires further study [KB].

1.1. Our Contribution

In this work, we construct a new lattice based PCS, Jindo¹, optimized for client-side proving. We choose lattice cryptography because several recent constructions [HSS24, NS24, KLN⁺26] exhibit faster proof generation than other approaches, which is the key priority in client-side proving. Specifically, we observe that the two recent constructions CELPC [HSS24] and Greyhound [NS24] have strong potential for client-side proving, but still have several limitations that hinder practical deployment. We therefore build a new lattice based PCS that addresses these limitations through the following contributions.

Flexible Parameters. First, we design our PCS to support flexible parameter regimes so that it can be deployed in various client-side proving scenarios. Specifically, we observe that CELPC supports this feature by introducing a novel encoding method tailored to the lattice setting. This allows a wide range of prime field sizes, from 128 to 1024 bits, which are particularly useful for proving statements about client-side ciphertexts or public keys in homomorphic encryption based protocols [HMSS25a, HMSS25b, HLSS25]. However, a key limitation of CELPC is its large proof size, which stems from the structure of its polynomial evaluation protocol. More precisely, its evaluation protocol supports only a polynomial sized challenge set, which requires several repetitions to attain negligible knowledge soundness error. In contrast, the evaluation protocol in Greyhound supports an exponentially large challenge set based on the analysis of [LS18], which eliminates the need for repetition. Furthermore, it allows composition with the Labrador [BS23] proof system to further reduce proof size. However, its limitation lies in its restricted parameter regime, where the commitment modulus must equal the coefficient modulus of the input polynomials and be congruent to 5 modulo 8. This constraint prevents efficient polynomial multiplication via the number theoretic transform (NTT), which degrades overall performance. Thus, Greyhound efficiently supports only small prime fields, such as 32-bit prime fields, making it difficult to deploy in more general settings.

In Jindo, we resolve these parameter-related issues by generalizing the evaluation protocol of Greyhound to support a Greyhound-like extraction method while allowing a CELPC-like encoding method without imposing constraints on the commitment modulus. As a result, the evaluation protocol of Jindo supports flexible parameter regimes while offering a large challenge set, which avoids large proof

sizes by eliminating repetitions. Moreover, we can choose NTT-friendly commitment moduli to enhance prover performance.

Sublinear Masking. Next, we design our PCS to achieve evaluation hiding with small computational overhead. Evaluation hiding is crucial for achieving zero-knowledgeness in PIOP-based proof systems, but it often introduces non-trivial prover overhead. One way to measure this overhead is to compare the size of the additional masking values required for evaluation hiding with the input polynomial size N , as proposed in [CFW26]. For example, a common approach to achieving evaluation hiding is to generate commitments to masking values of size N and use them to hide the witness through batched evaluation, as in [BDFG21, GLH⁺25, LZC⁺26]. However, this approach generally requires $O(N)$ masking overhead, which makes proof generation at least twice as slow. Although CELPC and Greyhound achieve sublinear masking overhead without using masking commitments, their techniques significantly degrade prover performance. More precisely, CELPC achieves evaluation hiding by randomizing the entire input witness itself, which does not generate masking commitments but incurs $O(N)$ additional computational overhead. In contrast, evaluation hiding in Greyhound relies on the zero-knowledge version of Labrador, which has no concrete implementation, so its actual performance overhead is unknown.

In Jindo, we achieve evaluation hiding with sublinear masking overhead by generalizing the masking commitment technique. We observe that the overhead of masking commitments is amortized as the number of batched polynomials increases, since a single masking commitment can cover all of them. To exploit this property, we adapt the subpolynomial evaluation structure of [LZC⁺26], which was originally introduced for distributed SNARK generation. Specifically, our PCS first splits the input polynomial into m subpolynomials and generates a commitment for each subpolynomial. The evaluation protocol then reconstructs the evaluation of the input polynomial through batched evaluation of these subpolynomials. As a result, we reduce the masking overhead from $O(N)$ to $O(N/m)$, achieving sublinear masking overhead depending on the choice of m .

Overall Complexity Improvement. We further observe that the choice of the number m of split subpolynomials determines not only the masking overhead for evaluation hiding, but also the proof size and verification complexity. More precisely, since our evaluation protocol is based on the construction of Greyhound, whose proof size and verification complexity are square root in the input size², our PCS yields complexity $\tilde{O}(m + \sqrt{N/m})$ for both proof size and verification. Here, the $\tilde{O}(m)$ term comes from aggregating the commitments for the subpolynomials in batched evaluation, while the $\tilde{O}(\sqrt{N/m})$ term comes from applying the evaluation protocol to the aggregated subpolynomial of size

1. Named after the Jindo dog breed, following the naming convention of Greyhound and Labrador.

2. Here, we consider the evaluation protocol without composition with Labrador.

N/m . Therefore, we choose $m = O(N^{1/3})$, which yields a cube root complexity of $\tilde{O}(N^{1/3})$ for both proof size and verification, while achieving a sublinear masking overhead of $O(N^{2/3})$.

Implementation. Finally, we provide an implementation of our PCS to demonstrate its practical efficiency. To further enhance prover performance, which is our main design focus, we employ various optimization techniques, including SIMD-based acceleration, following recent lattice-based PCSs [BKS⁺21, KLN⁺26, NS24]. In the PCS benchmark, Jindo achieves about an order of magnitude improvement over CELPC across all metrics, including proof generation, verification, and proof size, thanks to the improved evaluation protocol in our PCS. When compared with the recent code-based PCS PIP-FRI [LZC⁺26], which also targets fast proving performance and evaluation hiding, Jindo still achieves about an order of magnitude improvement in proof generation while providing similar verification cost and proof size.

In addition to the PCS benchmark, we demonstrate the efficiency of Jindo in PIOP compilation scenarios. Specifically, we provide benchmark results for proving the validity of RLWE samples by compiling the PIOPs presented in [HLSS25], since this is one of the most commonly proven statements in various lattice-based protocols [BGL⁺23, ABGS23, AGJ⁺24, CHM⁺25, HMSS25a, HMSS25b, BCGL⁺25]. The benchmark results show that compilation with Jindo achieves about an order of magnitude improvement over compilation with CELPC across all metrics, due to the performance improvement of our PCS. Beyond PIOP-based approaches, we also compare with the Lazer library [LSS24], which is dedicated to proving lattice relations. The results show that Jindo achieves about an order of magnitude improvement in proof generation and about two orders of magnitude improvement in verification, making Jindo a strong candidate for client-side proving scenarios.

1.2. Related Work

Lattice-based PCS. Apart from Greyhound and CELPC, whose security relies on standard lattice problems, another line of work explores PCS constructions based on structured short integer solution assumptions [ACL⁺22, FLV23, KLNO24, KLNO25, KLN⁺26]. These constructions primarily aim to reduce verification complexity to a poly-logarithmic scale by leveraging additional structure in the underlying assumptions. Notably, the most recent construction [KLN⁺26] achieves practical performance in prover time, verification time, and proof size. However, the underlying cryptographic assumptions require further study, and evaluation hiding remains unaddressed.

Code-based PCS. Code-based constructions are popular alternatives for post-quantum PCS. Several works [LZC⁺26, GLH⁺25, ACFY25, ZXZS20] study PCS constructions based on Reed-Solomon codes. Notably, some of these constructions [ZXZS20, LZC⁺26, GLH⁺25] provide explicit

methods for achieving evaluation hiding. Another line of work [GLS⁺23, ZCF24, BCF⁺25, DP25] studies constructions based on other error-correcting codes to obtain field-agnostic compatibility, which is useful for certain PIOP applications, or faster proving performance over binary extension fields. However, evaluation hiding in these constructions remains unknown.

2. Technical Overview

In this section, we provide a technical overview of our PCS construction. At a high level, the evaluation protocol in our PCS can be interpreted as a batched quadratic form evaluation protocol, where each quadratic form corresponds to a subpolynomial split from an input polynomial. Throughout this section, we focus on the evaluation of multilinear polynomials over a prime field $\mathbb{Z}_p$ for simplicity. The more general setting, which includes support for extension fields, is presented in Section 4.

2.1. Ajtai Commitment with CELPC Encoding

We first review the variant of the Ajtai commitment scheme introduced in CELPC to incorporate its encoding method, which efficiently supports commitments to coefficients over a large prime field $\mathbb{Z}_p$. The scheme is defined over the cyclotomic ring $R = \mathbb{Z}[X]/(X^d + 1)$, where d is a power of two, and has the following form.

$$\mathbf{t} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{r} \pmod{q}.$$

Here, the commitment keys $\mathbf{A} \in R_q^{\mu \times m}$ and $\mathbf{B} \in R_q^{\mu \times (\mu + \nu)}$ are uniformly random matrices over $R_q = R/qR$. The vector $\mathbf{x} \in R^m$ encodes the input message, while $\mathbf{r} \in R^{\mu + \nu}$ denotes the randomness used during commitment generation. The resulting commitment $\mathbf{t} \in R_q^\mu$ is a vector over R_q . The binding property is based on the hardness of the Module Short Integer Solution (MSIS) problem, and its security depends on the parameter μ and the norm bounds on $\mathbf{x}$ and $\mathbf{r}$, which should be smaller than q . The hiding property follows from the hardness of the Module Learning with Errors (MLWE) problem, and its security is determined by the parameter ν . A key feature of the Ajtai commitment scheme is its compressing property, where the commitment dimension μ can be chosen as $\tilde{O}(1)$ with respect to the message dimension m .

Encoding for Large Prime Fields. Due to the norm constraints required for MSIS security, coefficients in $\mathbb{Z}_p$ cannot be committed directly when $p \gg q$. To address this issue, CELPC introduces an encoding method derived from the isomorphism $\mathbb{Z}_p^\gamma \cong R_{X^\gamma - b} = R/(X^\gamma - b)R$, which holds when $p = b^{d/\gamma} + 1$ for $\gamma \mid d$. The encoding map is then defined as follows.

$$\text{Ecd} : \mathbf{a} = (a_0, \dots, a_{\gamma-1}) \in \mathbb{Z}_p^\gamma \mapsto \hat{a}(X) = \sum_{i=0}^{\gamma-1} \sum_{j=0}^{d/\gamma-1} a_{i,j} X^{i+\gamma j} \in R$$

Here, each coefficient a_i is represented as $a_i = \sum_{j=0}^{d/\gamma-1} a_{i,j} b^j$, where each digit $a_{i,j}$ is bounded by $b \ll q$. Consequently, a vector $\mathbf{a} \in \mathbb{Z}_p^\gamma$ is encoded as a ring element

$\hat{a} \in R$ whose norm remains much smaller than q , allowing it to be committed using the Ajtai commitment scheme. As an abuse of notation, for a single element $\alpha \in \mathbb{Z}_p$, we define $\text{Ecd}(\alpha) := \text{Ecd}(\alpha, 0, \dots, 0)$. Then, for any $\mathbf{a}, \mathbf{b} \in \mathbb{Z}_p^\gamma$, the following linear homomorphism holds.

$$\text{Ecd}(\alpha) \cdot \text{Ecd}(\mathbf{a}) + \text{Ecd}(\mathbf{b}) = \text{Ecd}(\alpha \cdot \mathbf{a} + \mathbf{b}) \pmod{X^\gamma - b}$$

Outer Commitment with Rounding Compression. Since our evaluation protocol is based on quadratic forms, we need to commit to a matrix $\hat{\mathbf{F}} \in R^{m \times n}$ using the Ajtai commitment scheme. However, committing to all entries with a single Ajtai commitment does not provide an efficient evaluation structure. We therefore use the outer commitment structure of Greyhound, where each column is first committed using an Ajtai commitment, and these commitments are then committed again using another Ajtai commitment to further reduce the commitment size.

More precisely, the prover first commits to each column $\hat{\mathbf{f}}_k$ of $\hat{\mathbf{F}}$ as $\mathbf{t}_k := \mathbf{A}\hat{\mathbf{f}}_k + \mathbf{B}\mathbf{r}_k \pmod{q}$. Rounding is then applied to each $\mathbf{t}_k$ as $\hat{\mathbf{t}}_k := \lfloor \mathbf{t}_k / B_t \rfloor \in R^\mu$ for some $B_t > 0$, so that they can be committed again. A final Ajtai commitment $\mathbf{u} \in R_{q_o}^\kappa$ is then generated as $\mathbf{u} := \mathbf{D}(\hat{\mathbf{t}}_0 \parallel \dots \parallel \hat{\mathbf{t}}_{n-1}) \pmod{q_o}$ for a random matrix $\mathbf{D} \in R_{q_o}^{\kappa \times \mu n}$. Finally, rounding is further applied to $\mathbf{u}$ as $\hat{\mathbf{u}} := \lfloor \mathbf{u} / B_u \rfloor$ for some $B_u > 0$ to reduce the commitment size. We refer to each $\hat{\mathbf{t}}_k \in R^\mu$ as an inner commitment and to $\hat{\mathbf{u}} \in R^\kappa$ as an outer commitment. The outer commitment $\hat{\mathbf{u}}$ then serves as the commitment to $\hat{\mathbf{F}}$.

2.2. Quadratic Form Evaluation Protocol

Based on the outer commitment structure, we build a quadratic form evaluation protocol that serves as the core building block for our polynomial evaluation protocol.

Single Quadratic Form Evaluation. The central algebraic relation that we aim to prove is knowledge of a matrix $\hat{\mathbf{F}} \in R^{m \times n}$ consisting of short elements that satisfies the encoded quadratic form relation $\hat{\mathbf{v}}^\top \hat{\mathbf{F}} \hat{\mathbf{w}} = \hat{y} \pmod{X^\gamma - b}$. Here, $\hat{\mathbf{v}} \in R^m$ and $\hat{\mathbf{w}} \in R^n$ are public encoded vectors, and $\hat{y} \in R_{X^\gamma - b}$ is the claimed encoded output. To prove this relation, we first commit to $\hat{\mathbf{F}}$ using the outer commitment structure, which yields the outer commitment $\hat{\mathbf{u}}$. The evaluation protocol then proceeds as follows.

In the first round, the prover sends the partial evaluation $\mathbf{z} \in R^n$ and the concatenated inner commitments $\hat{\mathbf{t}} \in R^{\mu n}$, defined by $\mathbf{z}^\top := \hat{\mathbf{v}}^\top \hat{\mathbf{F}}$ and $\hat{\mathbf{t}} := \hat{\mathbf{t}}_0 \parallel \dots \parallel \hat{\mathbf{t}}_{n-1}$. The verifier then samples a challenge vector $\mathbf{c} = (c_0, \dots, c_{n-1}) \leftarrow \mathcal{C}^n$, where $\mathcal{C} \subset R$ consists of short elements. After receiving the challenge, the prover responds with

$$\bar{\mathbf{f}} := \hat{\mathbf{F}}\mathbf{c} = \sum_{k=0}^{n-1} c_k \hat{\mathbf{f}}_k, \quad \bar{\mathbf{r}} := \mathbf{R}\mathbf{c} = \sum_{k=0}^{n-1} c_k \mathbf{r}_k,$$

where $\mathbf{R} = [\mathbf{r}_0 \mid \dots \mid \mathbf{r}_{n-1}] \in R^{(\mu+\nu) \times n}$. Finally, the verifier checks the norm constraints and algebraic relations required for $\bar{\mathbf{f}}$ and $\bar{\mathbf{r}}$.

For extraction, we use coordinate-wise special soundness [FMN24, BBC⁺18], as in Greyhound. However, our

evaluation protocol requires $c, c' \xleftarrow{\$} \mathcal{C}$ to satisfy that $c - c'$ is invertible in the message space $R_{X^\gamma - b}$ with overwhelming probability, rather than in the commitment space R_q . Thus, our evaluation protocol improves upon Greyhound by decoupling the commitment modulus q from the coefficient modulus p , which makes the protocol compatible with the CELPC encoding while allowing an exponentially large challenge set that avoids repetition. Both the communication and verification complexities are $\tilde{O}(m+n)$, matching those of Greyhound.

Batched Quadratic Form Evaluation. Next, we describe the batched quadratic form evaluation protocol, which yields the main complexity improvement in our PCS. Suppose that we evaluate multiple quadratic forms $\hat{\mathbf{v}}^\top \hat{\mathbf{F}}_i \hat{\mathbf{w}} = \hat{y}_i$ for $0 \leq i < t$, and let $\hat{\mathbf{u}}_i$ denote the outer commitment for each $\hat{\mathbf{F}}_i$. For batched evaluation, the prover and verifier aggregate these quadratic forms into a single quadratic form through a random linear combination, and then run the evaluation protocol for the aggregated quadratic form.

More precisely, for a random challenge $\boldsymbol{\alpha} := (\alpha_0, \dots, \alpha_{t-1}) \xleftarrow{\$} \mathcal{C}^t$ sampled by the verifier, the prover and verifier run the quadratic form evaluation protocol for $\hat{\mathbf{v}}^\top \hat{\mathbf{F}} \hat{\mathbf{w}} = \hat{y}$ with the outer commitment $\hat{\mathbf{u}}$, where

$$\hat{\mathbf{F}} := \sum_{i=0}^{t-1} \alpha_i \cdot \hat{\mathbf{F}}_i, \quad \hat{\mathbf{u}} := \sum_{i=0}^{t-1} \alpha_i \cdot \hat{\mathbf{u}}_i, \quad \hat{y} := \sum_{i=0}^{t-1} \alpha_i \cdot \hat{y}_i.$$

This aggregation allows extraction of each $\hat{\mathbf{F}}_i$ based on coordinate wise special soundness with respect to the random challenge $\boldsymbol{\alpha}$. Therefore, the batched evaluation protocol has complexity $\tilde{O}(m+n+t)$ for evaluating t quadratic forms. Here, the $\tilde{O}(t)$ term accounts for aggregating $\hat{\mathbf{u}} = \sum_{i=0}^{t-1} \alpha_i \cdot \hat{\mathbf{u}}_i$ and $\hat{y} = \sum_{i=0}^{t-1} \alpha_i \cdot \hat{y}_i$, while the $\tilde{O}(m+n)$ term accounts for evaluating $\hat{\mathbf{v}}^\top \hat{\mathbf{F}} \hat{\mathbf{w}} = \hat{y}$. This improves over the naive approach, which requires $\tilde{O}(t(m+n))$.

2.3. Polynomial Evaluation Protocol

We now construct a polynomial evaluation protocol based on the batched quadratic form evaluation protocol. Let $f \in \mathbb{Z}_p^{<2}[X_0, \dots, X_{D-1}]$ be a multilinear polynomial in D variables, where $2^D = N = mn$. Following the approach in [WTS⁺18], we index each monomial in $\mathbb{Z}_p^{<2}[X_0, \dots, X_{D-1}]$ as $\tilde{X}_i := \prod_{k=0}^{D-1} X_k^{i_k}$ for the binary representation $(i_0, \dots, i_{D-1})_2$ of $0 \leq i < N$. Then, the evaluation at a point $\mathbf{x} \in \mathbb{Z}_p^D$ can be represented as the following quadratic form.

$$y = f(\mathbf{x}) = \mathbf{v}^\top \mathbf{F} \mathbf{w} \pmod{p},$$

where the (i, j) -th entry of $\mathbf{F} \in \mathbb{Z}_p^{m \times n}$ corresponds to the $(n \cdot i + j)$ -th coefficient of f , and we define $\mathbf{v} = (1, \tilde{x}_n, \dots, \tilde{x}_{(m-1)n})$ and $\mathbf{w} = (1, \tilde{x}_1, \dots, \tilde{x}_{n-1})$. To incorporate batched quadratic form evaluation and the CELPC encoding method simultaneously, we decompose $\mathbf{v}$ and $\mathbf{w}$ into tensor products as follows.

$$\mathbf{v}_0 := (1, \tilde{x}_{m_1 n}, \dots, \tilde{x}_{(m_0-1)m_1 n}), \quad \mathbf{v}_1 := (1, \tilde{x}_n, \dots, \tilde{x}_{(m_1-1)n}), \\ \mathbf{w}_0 := (1, \tilde{x}_{n_1}, \dots, \tilde{x}_{(n_0-1)n_1}), \quad \mathbf{w}_1 := (1, \tilde{x}_1, \dots, \tilde{x}_{n_1-1}),$$

where $m = m_0 m_1$ and $n = n_0 n_1$, so that $\mathbf{v} = \mathbf{v}_0 \otimes \mathbf{v}_1$ and $\mathbf{w} = \mathbf{w}_0 \otimes \mathbf{w}_1$. Next, we interpret the coefficient vector of f as a tensor indexed by $(i, j, k, \ell) \in [m_0] \times [m_1] \times [n_0] \times [n_1]$. The first dimension indexes the batch, and the last dimension is used for the CELPC encoding. Then, the evaluation $y = f(\mathbf{x})$ can be rewritten as

$$y = f(\mathbf{x}) = (\mathbf{v}_0 \otimes \mathbf{v}_1)^\top \mathbf{F}(\mathbf{w}_0 \otimes \mathbf{w}_1) \\ = \sum_{i=0}^{m_0-1} \sum_{j=0}^{m_1-1} \sum_{k=0}^{n_0-1} \sum_{\ell=0}^{n_1-1} f_{i,j,k,\ell} \cdot v_{0,i} v_{1,j} w_{0,k} w_{1,\ell}$$

where $f_{i,j,k,\ell}$ denotes the $(m_1 n_0 n_1 i + n_0 n_1 j + n_1 k + \ell)$ -th coefficient of f . This follows from the relation $\tilde{x}_{m_1 n_0 n_1 i + n_0 n_1 j + n_1 k + \ell} = v_{0,i} v_{1,j} w_{0,k} w_{1,\ell}$, which holds for each i, j, k, ℓ . Based on this evaluation structure, we build a polynomial evaluation protocol over $\mathbb{Z}_p$ using batched quadratic form evaluation over $R_{X^\gamma-b}$ as follows.

Commitment Generation. First, we explain how the coefficients of the input polynomial f are encoded into matrices over R for batched quadratic form evaluation. We begin by splitting f into subpolynomials $f_0, \dots, f_{m_0-1}$, each with N/m_0 coefficients, following the decomposition $\mathbf{v} = \mathbf{v}_0 \otimes \mathbf{v}_1$. More precisely, each f_i is defined as

$$f_i(\mathbf{X}) = \sum_{j=0}^{m_1-1} \sum_{k=0}^{n_0-1} \sum_{\ell=0}^{n_1-1} f_{i,j,k,\ell} \cdot \tilde{X}_{n_0 n_1 j} \tilde{X}_{n_1 k} \tilde{X}_\ell$$

so that $f(\mathbf{X}) = \sum_{i=0}^{m_0-1} f_i(\mathbf{X}) \cdot \tilde{X}_{m_1 n_0 n_1 i}$. Next, we encode each subpolynomial f_i into a matrix $\hat{\mathbf{F}}_i$ through the CELPC encoding, based on the decomposition $\mathbf{w} = \mathbf{w}_0 \otimes \mathbf{w}_1$. Specifically, let $\mathbf{f}_{i,j,k} := (f_{i,j,k,0}, \dots, f_{i,j,k,n_1-1}) \in \mathbb{Z}_p^{n_1}$ and let $\hat{\mathbf{f}}_{i,j,k} := \text{Ecd}(\mathbf{f}_{i,j,k})$ by setting $n_1 = \gamma$. Then, the coefficient matrix of f_i is encoded as $\hat{\mathbf{F}}_i \in R^{m_1 \times n_0}$, whose (j, k) -th entry is $\hat{\mathbf{f}}_{i,j,k}$. Finally, each outer commitment $\hat{\mathbf{u}}_i$ is generated from $\hat{\mathbf{F}}_i$, which encodes all coefficients of f_i .

Evaluation Protocol. We now describe how the proof of the polynomial evaluation $y = f(\mathbf{x})$ can be achieved through the batched quadratic form evaluations of $\hat{\mathbf{F}}_i$ for $0 \leq i < m_0$. Suppose that the outer commitment $\hat{\mathbf{u}}_i$ is generated for each $\hat{\mathbf{F}}_i$. In the first round, the prover sends $\hat{y}_0, \dots, \hat{y}_{m_0-1} \in R_{X^\gamma-b}$, where $\hat{y}_i := \hat{\mathbf{v}}_1^\top \hat{\mathbf{F}}_i \hat{\mathbf{w}}_0 \pmod{X^\gamma - b}$, with the encoded public vectors $\hat{\mathbf{v}}_1 := (\text{Ecd}(v_{1,0}), \dots, \text{Ecd}(v_{1,m_1-1})) \in R^{m_1}$ and $\hat{\mathbf{w}}_0 := (\text{Ecd}(w_{0,0}), \dots, \text{Ecd}(w_{0,n_0-1})) \in R^{n_0}$. Upon receiving these values, the verifier checks

$$\sum_{i=0}^{m_0-1} \langle \text{Dcd}(\hat{y}_i), \mathbf{w}_1 \rangle \cdot v_{0,i} \stackrel{?}{=} y \pmod{p}.$$

This check ensures $y = f(\mathbf{x})$, since

$$\langle \text{Dcd}(\hat{y}_i), \mathbf{w}_1 \rangle = \sum_{j,k,\ell} f_{i,j,k,\ell} \cdot v_{1,j} w_{0,k} w_{1,\ell} = f_i(\mathbf{x})$$

and $f(\mathbf{x}) = \sum_{i=0}^{m_0-1} f_i(\mathbf{x}) \cdot v_{0,i}$. Thus, the polynomial relation $y = f(\mathbf{x})$ is reduced to the quadratic form relations $\hat{\mathbf{v}}_1^\top \hat{\mathbf{F}}_i \hat{\mathbf{w}}_0 \stackrel{?}{=} \hat{y}_i \pmod{X^\gamma - b}$ for $0 \leq i < m_0$. Finally, the prover and verifier invoke the batched quadratic form evaluation protocol on these quadratic forms, which completes the polynomial evaluation protocol.

Since the proof size and verification complexity of the batched evaluation protocol scale as $\tilde{O}(m_0 + m_1 + n_0)$, the protocol achieves balanced complexity $\tilde{O}(N^{1/3})$ when m_0 , m_1 , and n_0 are balanced, with $n_1 = \gamma$ treated as a constant, since $N = m_0 m_1 n_0 n_1$. We also note that our polynomial evaluation method naturally supports the univariate case by setting $\tilde{X}_i = X^i$ for $0 \leq i < N$.

2.4. Achieving Evaluation Hiding

We note that the polynomial evaluation protocol defined above does not yet achieve evaluation hiding. Specifically, the values $\hat{y}_i$ arising from the subpolynomials f_i reveal more information about f than the target evaluation $y = f(\mathbf{x})$. In addition, the values $\mathbf{z}$, $\bar{\mathbf{f}}$, and $\bar{\mathbf{r}}$ that appear in the final quadratic form evaluation protocol after aggregation are not simulatable. We resolve these issues by devising a new evaluation hiding technique tailored to our evaluation protocol, while minimizing the additional prover overhead.

Splitting with Augmented Subpolynomial. To resolve the leakage from subpolynomial evaluations, we devise an augmented subpolynomial structure that introduces random auxiliary polynomials g_i when generating the outer commitments. More precisely, each matrix $\hat{\mathbf{F}}_i$ is augmented as

$$\hat{\mathbf{F}}_i^* := [\hat{\mathbf{f}}_{i,0}^* \mid \dots \mid \hat{\mathbf{f}}_{i,n_0-1}^*] = \begin{bmatrix} \hat{\mathbf{F}}_i \\ \hat{\mathbf{g}}_i^\top \end{bmatrix} \in R^{(m_1+1) \times n_0},$$

where $\hat{\mathbf{f}}_{i,k}^* = (\hat{\mathbf{f}}_{i,k}, \hat{g}_{i,k}) \in R^{m_1+1}$ for a random sample $\hat{\mathbf{g}}_i = (\hat{g}_{i,0}, \dots, \hat{g}_{i,n_0-1}) \stackrel{\$}{\leftarrow} R_{X^\gamma-b}^{n_0}$. Then, each commitment represents an augmented subpolynomial $f_i^*(\mathbf{X}, X^*) := f_i(\mathbf{X}) + X^* \cdot g_i(\mathbf{X})$, where g_i is derived from $\hat{\mathbf{g}}_i$ for $0 \leq i < m_0$. We also define $g(\mathbf{X}) := \sum_{i=0}^{m_0-1} g_i(\mathbf{X}) \cdot \tilde{X}_{N i / m_0}$. Based on this augmented subpolynomial structure, we construct a splitting protocol that reduces the input polynomial evaluation relation to multiple quadratic form relations while achieving simulatability.

In the first round, the prover sends the value

$$y_g := g(\mathbf{x}) = \sum_{i=0}^{m_0-1} g_i(\mathbf{x}) \cdot \tilde{x}_{N i / m_0} = \sum_{i=0}^{m_0-1} g_i(\mathbf{x}) \cdot v_{0,i} \pmod{p}.$$

The verifier then samples a challenge $x^* \stackrel{\$}{\leftarrow} \mathbb{Z}_p^\times$. Upon receiving the challenge x^* , the prover responds with $\hat{y}_i^* \in R_{X^\gamma-b}$ for $0 \leq i < m_0$, defined as

$$\hat{y}_i^* := \hat{\mathbf{v}}_1^\top \hat{\mathbf{F}}_i^* \hat{\mathbf{w}}_0 = \hat{y}_i + x^* \cdot \hat{\mathbf{g}}_i^\top \hat{\mathbf{w}}_0 \pmod{X^\gamma - b},$$

where $\hat{\mathbf{v}}_1^* := (\hat{\mathbf{v}}_1, \text{Ecd}(x^*)) \in R^{m_1+1}$. Finally, the verifier checks

$$y + x^* \cdot y_g \stackrel{?}{=} \sum_{i=0}^{m_0-1} \langle \text{Dcd}(\hat{y}_i^*), \mathbf{w}_1 \rangle \cdot v_{0,i} \pmod{p}.$$

In this protocol, simulatability is achieved by sending $\hat{y}_i^*$ instead of $\hat{y}_i$. More precisely, each $\hat{y}_i^*$ appears random because $\hat{y}_i$ is masked by $x^* \cdot \hat{\mathbf{g}}_i^\top \hat{\mathbf{w}}_0$, where $\hat{\mathbf{g}}_i$ is sampled uniformly from $R_{X^\gamma-b}^{n_0}$ and x^* is invertible. For soundness, the verifier's check implies $y + x^* \cdot y_g = f(\mathbf{x}) + x^* \cdot g(\mathbf{x}) \pmod{p}$. Therefore, if $y \neq f(\mathbf{x})$, the verifier accepts only

if a nonzero linear polynomial in x^* vanishes. By the Schwartz-Zippel lemma, this occurs with probability at most $1/(p-1)$, which is negligible when p is exponential in the security parameter.

Aggregation with Rejection Sampling. Next, we enhance the aggregation procedure so that the subsequent quadratic form evaluation protocol attains simulatability. To achieve this, we hide the aggregated values by introducing a masking outer commitment together with the rejection sampling method of [Lyu12]. More precisely, in the first round, the prover generates a masking outer commitment $\hat{\mathbf{u}}_{m_0}$ using $\hat{\mathbf{F}}_{m_0}^*$ and $\mathbf{R}_{m_0}$, which are sampled from a discrete Gaussian distribution. The prover then sends $\hat{\mathbf{u}}_{m_0}$ together with $\hat{y}_{m_0}^* := \hat{\mathbf{v}}_1^{\top} \hat{\mathbf{F}}_{m_0}^* \hat{\mathbf{w}}_0 \pmod{X^\gamma - b}$. The verifier then sends a random challenge $\alpha = (\alpha_0, \dots, \alpha_{m_0-1}) \xleftarrow{\$} \mathcal{C}^{m_0}$. After receiving the challenge, the prover computes $\hat{\mathbf{F}}^* := \sum_{i=0}^{m_0-1} \alpha_i \hat{\mathbf{F}}_i^* + \hat{\mathbf{F}}_{m_0}^*$ and $\mathbf{R} := \sum_{i=0}^{m_0-1} \alpha_i \mathbf{R}_i + \mathbf{R}_{m_0}$, and then applies a rejection sampling procedure to these values. If the rejection procedure accepts, the prover and the verifier runs quadratic form evaluation protocol for the aggregated quadratic form relation $\hat{y}^* := \hat{\mathbf{v}}_1^{\top} \hat{\mathbf{F}}^* \hat{\mathbf{w}}_0$, where $\hat{y}^* := \sum_{i=0}^{m_0-1} \alpha_i \cdot \hat{y}_i^* + \hat{y}_{m_0}^*$.

This enhanced aggregation ensures simulatability because, by the rejection sampling procedure, the distribution of $(\hat{\mathbf{F}}^*, \mathbf{R})$ becomes statistically independent of the values of $\hat{\mathbf{F}}_i^*$ and $\mathbf{R}_i$ for $0 \leq i < m_0$. We also note that the additional overhead for generating the masking outer commitment is $O(m_1 n_0 n_1) = O(N^{2/3})$, which achieves sublinear masking overhead.

3. Preliminaries

3.1. Notation

For a positive integer q , we use $\mathbb{Z} \cap (-q/2, q/2]$ as a representative set of $\mathbb{Z}_q$ and denote by $[a]_q$ the reduction of a modulo q . We treat all vectors as column vectors and use $\|$ to denote the concatenation of two vectors. Let d be a power of two. We denote by $R = \mathbb{Z}[X]/(X^d + 1)$ the ring of integers of the $2d$ -th cyclotomic field and by $R_q = \mathbb{Z}_q[X]/(X^d + 1)$ the residue ring of R modulo q . For a vector $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{Z}^n$, the L^p ($p \geq 1$) and L^∞ norms are defined as $\|\mathbf{v}\|_p := \sqrt[p]{\sum_{i=1}^n |v_i|^p}$ and $\|\mathbf{v}\|_\infty := \max_{1 \leq i \leq n} |v_i|$. We denote the Hadamard product by $\odot$ and the Kronecker product by $\otimes$. By abuse of notation, we often treat a polynomial as its coefficient vector. For example, for a polynomial $f \in R$, $\|f\|_p$ is defined over its coefficient vector. Additionally, for a distribution $\mathcal{D}$ over $\mathbb{Z}$, we denote $f \xleftarrow{\$} \mathcal{D}^d$ when each coefficient is sampled independently from $\mathcal{D}$.

3.2. Rejection Sampling

We review the rejection sampling method of [Lyu12], which is used to achieve simulatability in lattice-based sigma protocols.

Definition 1 (Discrete Gaussian). *The discrete Gaussian distribution over $\mathbb{Z}$ with standard deviation $\sigma > 0$ is defined as*

$$\mathcal{D}_\sigma(x) := \frac{\exp(-x^2/(2\sigma^2))}{\sum_{v \in \mathbb{Z}} \exp(-v^2/(2\sigma^2))}.$$

Lemma 1 (Lemma 4.4 [Lyu12]).

- 1) For any $k > 0$, $\Pr[|z| > k\sigma \mid z \xleftarrow{\$} \mathcal{D}_\sigma] \leq 2e^{-k^2/2}$.
- 2) For any $k > 1$, $\Pr[\|\mathbf{z}\|_2 > k\sigma\sqrt{m} \mid \mathbf{z} \xleftarrow{\$} \mathcal{D}_\sigma^m] < k^m e^{m(1-k^2)/2}$.

Lemma 2 (Lemma 2.4 [LNP22]). *Let $V \subset R^\ell$ be a set of polynomials with L^2 norm bound T , and let $\mathcal{V}$ be a probability distribution over V . Let Rej be the algorithm defined in Fig. 1 with $\sigma \approx \frac{14}{\ln M} T$ and $M > 0$. Then, the probability that $b = 0$ is at least $(1 - 2^{-128})/M$. Furthermore, the distribution $(\mathbf{v}, \mathbf{z})$ conditioned on $b = 0$ is within statistical distance $2^{-128}/M$ of the product distribution $\mathcal{V} \times \mathcal{D}_\sigma^{\ell d}$.*

$\text{Rej}(\mathbf{z}, \mathbf{v}, \sigma) \rightarrow 0/1$.

- 1) Sample $u \xleftarrow{\$} [0, 1)$.
- 2) If $u > \frac{1}{M} \cdot \exp\left(\frac{-2\langle \mathbf{z}, \mathbf{v} \rangle + \|\mathbf{v}\|_2^2}{2\sigma^2}\right)$, return 1.
- 3) Otherwise, return 0.

Figure 1: Rejection Sampling Algorithm

3.3. Challenge Set in Cyclotomic Rings

We introduce the notion of well-spreadness, which we use to describe challenge sets over cyclotomic rings.

Definition 2 (Well-Spreadness [BL26]). *Let $\mathcal{C} \subseteq R$, and suppose that $X^d + 1 = \prod_{i=0}^{\delta-1} (X^{d/\delta} - \zeta_i) \pmod{q}$ factors into δ irreducible factors in R_q . Then, $\mathcal{C}$ is called an η -well-spread set for R_q if, for every $0 \leq i < \delta$ and every $x \in \mathbb{Z}_q[X]/(X^{d/\delta} - \zeta_i)$, it holds that*

$$\Pr \left[c = x \pmod{X^{d/\delta} - \zeta_i} \mid c \xleftarrow{\$} \mathcal{C} \right] \leq \eta.$$

Lemma 3 (Lemma 34 [BL26], Lemma 1 [ESZ22]). *Let $R_q = \mathbb{Z}_q[X]/(X^d + 1)$, where $X^d + 1 = \prod_{i=0}^{\delta-1} (X^{d/\delta} - \zeta_i) \pmod{q}$ factors into δ irreducible factors in R_q . Let $\mathcal{S}_w \subseteq R$ be defined as*

$$\mathcal{S}_w := \left\{ c = \sum_{i=0}^{d/\delta-1} c_i X^i \in R \mid \begin{array}{l} \|c_i\|_\infty = 1 \wedge \|c_i\|_1 = w\delta/d \wedge \\ c_i \in \mathbb{Z}[X^{d/\delta}]/(X^d + 1) \end{array} \right\}.$$

Then, $\mathcal{S}_w$ is η -well-spread with

$$\eta = \max_{0 \leq i < \delta} \left(\frac{\beta}{q} + \frac{\beta}{q} \sum_{j=1}^{q-1} \left| \frac{1}{\delta} \sum_{\ell=0}^{\delta-1} \cos(2\pi j \zeta_i^\ell / q) \right|^{w\delta/d} \right)^{d/\delta} \quad (1)$$

where $\beta = \frac{\delta^{w\delta/d} (\delta - w\delta/d)!}{\delta!}$.

3.4. MSIS and MLWE

We review the hard problems in lattice cryptography that underlie our PCS.

Definition 3 (MSIS). Let μ, ℓ be positive integers, and let $0 < \beta < q$. We say that a PPT adversary A has advantage ε in solving MSIS $_{q,\mu,\ell,\beta}$ if

$$\Pr \left[[A \mid \mathbf{I}_\mu] \mathbf{x} = \mathbf{0} \pmod{q} \wedge \|\mathbf{x}\|_2 < \beta \mid A \xleftarrow{\$} R_q^{\mu \times \ell}, \mathbf{x} \leftarrow A(A) \right] \geq \varepsilon.$$

Definition 4 (MLWE). Let ν, ℓ be positive integers, and let χ be a distribution over R . We say that a PPT adversary A has advantage ε in solving MLWE $_{q,\nu,\ell,\chi}$ if

$$\left| \Pr \left[b = 1 \mid A \xleftarrow{\$} R_q^{\ell \times \nu}, \mathbf{r} \xleftarrow{\$} \chi^{\nu + \ell}, b \leftarrow A(A, [A \mid \mathbf{I}_\ell] \mathbf{r}) \right] - \Pr \left[b = 1 \mid A \xleftarrow{\$} R_q^{\ell \times \nu}, \mathbf{u} \xleftarrow{\$} R_q^\ell, b \leftarrow A(A, \mathbf{u}) \right] \right| \geq \varepsilon.$$

The parameter ℓ in both problems does not play a crucial role in determining their hardness, so we often omit it. To assess the hardness of the MSIS problem, we use the following bound from the security analysis of [GN08, MR09], where δ denotes the root Hermite factor.

$$\beta \leq \min \left\{ q, 2^{2\sqrt{\mu d \log q \log \delta}} \right\}. \quad (2)$$

3.5. Relaxed Reduction of Knowledge

We define the notion of relaxed reduction of knowledge, which simultaneously captures the soundness slack that arises in many lattice-based sigma protocols [BCK⁺14, LS18] and the reduction of knowledge framework [KP23], which generalizes interactive arguments of knowledge for protocol composition³. We also relax the completeness condition to incorporate rejection sampling, as in [BCK⁺14].

Definition 5 (Relaxed Reduction of Knowledge). Let $\Pi = (\text{Setup}, P, V)$ be an interactive protocol with the following syntax.

- $\text{Setup}(1^\lambda) \rightarrow \text{pp}$. Given a security parameter λ , output public parameters pp .
- $P(\text{pp}, x_1, w_1) \rightarrow (x_2, w_2)$. Given public parameters pp , a public instance x_1 , and a witness w_1 , output a reduced instance-witness pair (x_2, w_2) after interacting with V .
- $V(\text{pp}, x_1) \rightarrow x_2$. Given public parameters pp and a public instance x_1 , output a reduced instance x_2 after interacting with P .

The protocol Π is called a relaxed reduction of knowledge from a relation pair (R_1, R'_1) to a relation pair (R_2, R'_2) if $R_1 \subseteq R'_1$, $R_2 \subseteq R'_2$, and the following properties hold.

- **Completeness.** For every $(x_1, w_1) \in R_1$, the following holds for some failure rate $0 \leq \alpha < 1$.

$$\Pr \left[(x_2, w_2) \in R_2 \mid \begin{array}{c} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{tr}, x_2, w_2) \leftarrow \langle P(\text{pp}, x_1, w_1), V(\text{pp}, x_1) \rangle \end{array} \right] \geq 1 - \alpha.$$

- **Knowledge Soundness.** For every PPT adversary A and every deterministic polynomial-time prover P^* , there exists an expected PPT extractor E such that the following holds.

$$\Pr \left[\begin{array}{c} (x_2, w_2) \in R'_2 \wedge \\ (x_1, w'_1) \notin R'_1 \end{array} \mid \begin{array}{c} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{st}, x_1) \leftarrow A(\text{pp}) \\ (\text{tr}, x_2, w_2) \leftarrow \langle P^*(\text{pp}, \text{st}, x_1), V(\text{pp}, x_1) \rangle \\ w'_1 \leftarrow E^{P^*}(\text{pp}, \text{st}, x_1)(\text{pp}, x_1) \end{array} \right] = \text{negl}(\lambda).$$

3. An argument of knowledge can be viewed as a special case of a reduction of knowledge, where the reduced relation is the trivial relation $R_\top = \{(\text{true}, \perp)\}$, as noted in Definition 3 of [KP23].

- **Public Reducibility.** For every expected PPT adversary A and every deterministic polynomial-time prover P^* , there exists a deterministic polynomial-time function ϕ such that the following holds.

$$\Pr \left[\begin{array}{c} \phi(\text{pp}, x_1, \text{tr}) = x_2 \\ (\text{tr}, x_2, w_2) \leftarrow \langle P^*(\text{pp}, \text{st}, x_1), V(\text{pp}, x_1) \rangle \end{array} \mid \begin{array}{c} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{st}, x_1) \leftarrow A(\text{pp}) \end{array} \right] = 1.$$

The following lemma states that relaxed reductions of knowledge are closed under composition.

Lemma 4 (Theorem 1 [KP23]). Let Π_0 be a relaxed reduction of knowledge from (R_0, R'_0) to (R_1, R'_1) , and let Π_1 be a relaxed reduction of knowledge from (R_1, R'_1) to (R_2, R'_2) . Then, $\Pi_1 \circ \Pi_0$ is a relaxed reduction of knowledge from (R_0, R'_0) to (R_2, R'_2) .

3.6. Polynomial Commitment Scheme

We recall the definition of a polynomial commitment scheme, which is a special class of commitment scheme whose messages are polynomials and which supports evaluation proofs for committed polynomials.

Definition 6 (Polynomial Commitment). A polynomial commitment scheme is a tuple of PPT algorithms $(\text{Setup}, \text{Com}, \text{Open}, \text{Eval})$ with the following syntax.

- $\text{Setup}(1^\lambda, D) \rightarrow \text{pp}$. Given a security parameter λ and the number of variables D , output public parameters pp .
- $\text{Com}(\text{pp}, f) \rightarrow (\text{com}, \text{op})$. Given public parameters pp and a polynomial $f \in \mathbb{F}^{<2}[X_0, \dots, X_{D-1}]$, output a commitment com and an opening op .
- $\text{Open}(\text{pp}, \text{com}, f, \text{op}) \rightarrow 0/1$. Given public parameters pp , a commitment com , a polynomial $f \in \mathbb{F}^{<2}[X_0, \dots, X_{D-1}]$, and an opening op , output 0 or 1.
- $\text{Eval} = (\text{Eval.P}, \text{Eval.V})$. The evaluation protocol is a relaxed interactive argument of knowledge for a relation pair (R, R') . For an instance $x = (\text{com}, x, y)$ with $x \in \mathbb{F}^D$ and $y \in \mathbb{F}$, and a witness $w = (f, \text{op})$, the relaxed relation R' is defined as

$$R' := \left\{ (x, w) \mid \begin{array}{c} \text{Open}(\text{pp}, \text{com}, f, \text{op}) = 1 \wedge \\ f(x) = y \wedge f \in \mathbb{F}^{<2}[X_0, \dots, X_{D-1}] \end{array} \right\}. \quad (3)$$

Definition 7 (Binding). A polynomial commitment scheme satisfies binding if, for every PPT adversary A , the following holds.

$$\Pr \left[\begin{array}{c} f \neq f' \wedge \\ \text{Open}(\text{pp}, \text{com}, f, \text{op}) = 1 \wedge \\ \text{Open}(\text{pp}, \text{com}, f', \text{op}') = 1 \end{array} \mid \begin{array}{c} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (\text{com}, f, f', \text{op}, \text{op}') \leftarrow A(\text{pp}) \end{array} \right] \leq \text{negl}(\lambda).$$

Definition 8 (Hiding). A polynomial commitment scheme satisfies hiding if, for every PPT adversary A , the following holds.

$$\Pr \left[b = b' \mid \begin{array}{c} \text{pp} \leftarrow \text{Setup}(1^\lambda) \\ (f_0, f_1, \text{st}) \leftarrow A(\text{pp}) \\ b \xleftarrow{\$} \{0, 1\} \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f_b) \\ b' \leftarrow A(\text{st}, \text{com}) \end{array} \right] - \frac{1}{2} \leq \text{negl}(\lambda).$$

Definition 9 (Correctness). A polynomial commitment scheme $(\text{Setup}, \text{Com}, \text{Open}, \text{Eval})$ satisfies correctness if the

following holds for every $f \in \mathbb{F}^{<2}[X_0, \dots, X_{D-1}]$ and every $\mathbf{x} \in \mathbb{F}^D$.

$$\Pr \left[\begin{array}{c} (x, w) \in R \\ \text{pp} \leftarrow \text{Setup}(1^\lambda, D) \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f) \\ x := (\text{com}, \mathbf{x}, f(\mathbf{x})), w := (f, \text{op}) \end{array} \right] \geq 1 - \text{negl}(\lambda).$$

We define evaluation hiding following the definitions in [CHM⁺20, ZXZS20, LZC⁺26].

Definition 10 (Evaluation Hiding). *A polynomial commitment scheme $(\text{Setup}, \text{Com}, \text{Open}, \text{Eval})$ satisfies evaluation hiding if, for every PPT adversary $\mathcal{A}$, there exists a PPT simulator $\mathcal{S} = (\mathcal{S}_0, \mathcal{S}_1)$ such that the following holds for every $f \in \mathbb{F}^{<2}[X_0, \dots, X_{D-1}]$.*

$$\Pr \left[\begin{array}{c} \mathcal{A}(\text{pp}, \text{com}, \text{tr}) = 1 \wedge \\ b = 1 \\ \text{pp} \leftarrow \text{Setup}(1^\lambda, D) \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f) \\ \mathbf{x} \leftarrow \mathcal{A}(\text{pp}, \text{com}) \\ x := (\text{com}, \mathbf{x}, f(\mathbf{x})), w := (f, \text{op}) \\ (\text{tr}, b) \leftarrow (\text{Eval.P}(\text{pp}, x, w), \text{Eval.V}(\text{pp}, x)) \end{array} \right] - \Pr \left[\begin{array}{c} \mathcal{A}(\text{pp}, \text{com}, \text{tr}) = 1 \\ \text{pp} \leftarrow \text{Setup}(1^\lambda, D) \\ (\text{com}, \text{st}) \leftarrow \mathcal{S}_0(\text{pp}) \\ \mathbf{x} \leftarrow \mathcal{A}(\text{pp}, \text{com}) \\ x := (\text{com}, \mathbf{x}, f(\mathbf{x})) \\ \text{tr} \leftarrow \mathcal{S}_1(\text{pp}, \text{st}, x) \end{array} \right] \leq \text{negl}(\lambda).$$

4. Jindo Polynomial Commitment Scheme

In this section, we construct Jindo, a new lattice-based polynomial commitment scheme. We first review the CELPC encoding method and its generalization, which enables Jindo to support extension fields beyond prime fields. We then describe the commitment and opening algorithms for Jindo. Finally, we present our polynomial evaluation protocol, together with detailed security and complexity analyses. All proofs are deferred to Section B.

4.1. Encoding Method

We review the encoding method of CELPC and its recent generalization in [CHM⁺25, GV25], which additionally supports extension fields. The core idea of CELPC is to use the following algebraic structure

$$R_{X^\gamma-b} := \mathbb{Z}[X]/(X^d + 1, X^\gamma - b) \cong \mathbb{Z}_p[X]/(X^\gamma - b)$$

for $p = b^{d/\gamma} + 1$. This allows a vector in $\mathbb{Z}_p^\gamma$ to be encoded into a polynomial in $\mathbb{Z}_p[X]/(X^\gamma - b)$ by assigning the i -th entry to the i -th coefficient, viewing $\mathbb{Z}_p[X]/(X^\gamma - b)$ as a $\mathbb{Z}_p$ -vector space. Support for extension fields can be obtained by further analyzing the algebraic structure of $\mathbb{Z}_p[X]/(X^\gamma - b)$, as in [GV25], which gives the isomorphism $\mathbb{Z}_p[X]/(X^\gamma - b) = R_{X^\gamma-b} \cong \text{GF}(p^{d/\delta})^{\gamma\delta/d}$ when $p \equiv 2\delta + 1 \pmod{4\delta}$. We formalize this through two modes, coefficient encoding and slot encoding, which correspond to the original CELPC encoding and its generalized version, respectively.

Lemma 5 (Coefficient Encoding). *Let $p = b^{d/\gamma} + 1$ be a prime. Then, $R_{X^\gamma-b} \cong \mathbb{Z}_p^\gamma$ as a vector space under the coefficient isomorphism $\varphi_{\text{Coeff}} : \mathbb{Z}_p^\gamma \rightarrow R_{X^\gamma-b}$.*

Lemma 6 (Slot Encoding). *Let $p = b^{d/\gamma} + 1$ be a prime such that $p \equiv 2\delta + 1 \pmod{4\delta}$ for some $\delta \mid d$. Then, $R_{X^\gamma-b} \cong \text{GF}(p^{d/\delta})^{\gamma\delta/d}$ as a product of rings, with an isomorphism $\varphi_{\text{Slot}} : \text{GF}(p^{d/\delta})^{\gamma\delta/d} \rightarrow R_{X^\gamma-b}$.*

Based on the above two encoding methods, we present the encoding procedure for Jindo in Fig. 2, together with the flattening algorithm, which reduces the norm of encoded polynomials, and the decoding algorithm, which recovers a vector of field elements from an encoded polynomial.

- **Flatten**(a) $\rightarrow \hat{a}$: Given an input $a \in R_{X^\gamma-b}$, output a polynomial $\hat{a} \in R$ obtained as follows.
 - 1) Parse $a = \sum_{i=0}^{\gamma-1} a_i X^i \in \mathbb{Z}_p[X]/(X^\gamma - b)$ with $a_i \in \mathbb{Z}_p$.
 - 2) Write $a_i = (a_{i,d/\gamma-1} \dots a_{i,0})_b$ using the base- b representation.
 - 3) Output $\hat{a} = \sum_{i=0}^{\gamma-1} \sum_{j=0}^{d/\gamma-1} a_{i,j} \cdot X^{i+\gamma j} \in R$.
- **Ecd**($\mathbf{a}$) $\rightarrow \hat{a}$: Given a vector $\mathbf{a} \in \mathbb{F}^\rho$ and an isomorphism $\varphi \in \{\varphi_{\text{Coeff}}, \varphi_{\text{Slot}}\}$, return the polynomial $\hat{a} = \text{Flatten}(\varphi(\mathbf{a})) \in R$, where $\rho = \gamma$ if $\varphi = \varphi_{\text{Coeff}}$ and $\rho = \gamma\delta/d$ if $\varphi = \varphi_{\text{Slot}}$.
- **Ecd**(α) $\rightarrow \hat{a}$: Given an element $\alpha \in \mathbb{F}$, return the polynomial $\hat{a} = \text{Ecd}(\alpha \cdot \mathbf{1})$ if $\varphi = \varphi_{\text{Slot}}$ or the polynomial $\hat{a} = \text{Ecd}(\alpha, 0, \dots, 0)$ if $\varphi = \varphi_{\text{Coeff}}$.
- **Dcd**(a) $\rightarrow \mathbf{a}$: Given a polynomial $a \in R$ and an isomorphism $\varphi = \varphi_{\text{Coeff}}$ or $\varphi = \varphi_{\text{Slot}}$, return $\mathbf{a} = \varphi^{-1}(a \bmod (X^\gamma - b)) \in \mathbb{F}^\rho$.

Figure 2: Encoding Algorithm of Jindo

Below, we prove the coefficient bound for the flattening algorithm and the homomorphic property of the two encoding methods.

Lemma 7 (Flattening). *For every $a \in R_{X^\gamma-b}$, it holds that $\|\hat{a}\|_\infty \leq b$ and $\hat{a} = a \pmod{X^\gamma - b}$ for $\hat{a} = \text{Flatten}(a)$.*

Lemma 8 (Linear Homomorphism). *For both $\varphi = \varphi_{\text{Coeff}}$ and $\varphi = \varphi_{\text{Slot}}$, the following holds for all $\alpha \in \mathbb{F}$ and $\mathbf{v} \in \mathbb{F}^\rho$.*

$$\text{Ecd}(\alpha) \cdot \text{Ecd}(\mathbf{v}) = \text{Ecd}(\alpha \cdot \mathbf{v}) \pmod{X^\gamma - b}.$$

4.2. Scheme Description

We present the commitment and opening algorithms for Jindo as follows.

- **Setup**($1^\lambda, \rho$) $\rightarrow \text{pp}$: Given a security parameter λ and a structural parameter ρ , return public parameters pp defined as follows.
 - $q, q_0, B_t, B_u \in \mathbb{Z}_{>0}$.
 - $R = \mathbb{Z}[X]/(X^d + 1)$ and $p = b^{d/\gamma} + 1$.
 - $(\varphi, \mathbb{F}, n_1)$ is either $(\varphi_{\text{Coeff}}, \mathbb{Z}_p, \gamma)$ or $(\varphi_{\text{Slot}}, \text{GF}(p^{d/\delta}), \gamma\delta/d)$.
 - $N = mn$, $m = m_0 m_1$, and $n = n_0 n_1$.
 - $2^{D_0} = n_0$, $2^{D_1} = n_1$, $2^{D^*} = N/m_0$, and $2^D = N$.
 - $\mathbf{A} \xleftarrow{\$} R_q^{\mu \times (m_1+1)}$, $\mathbf{B} := [\mathbf{B}' \mid \mathbf{I}_\mu] \in R_q^{\mu \times (\mu+\nu)}$ for $\mathbf{B}' \xleftarrow{\$} R_q^{\mu \times \nu}$, and $\mathbf{D} \xleftarrow{\$} R_{q_0}^{\kappa \times \mu n_0}$.
 - $\mathcal{F} := \mathbb{F}^{<2}[X_0, \dots, X_{D-1}]$ and $\mathcal{F}^* := \mathbb{F}^{<2}[X_0, \dots, X_{D^*-1}] + X^* \cdot \mathbb{F}^{<2}[X_0, \dots, X_{D_0+D_1-1}]$.
 - $\tilde{X}_i := \prod_{k=0}^{D-1} X_k^{i_k}$ for $i = \sum_{k=0}^{D-1} i_k 2^k$.
 - $V_{0,i} := X_{m_1 n \cdot i}$ for $0 \leq i < m_0$ and $V_{1,j} := \tilde{X}_{n \cdot j}$ for $0 \leq j < m_1$.
 - $W_{0,k} := \tilde{X}_{n_1 \cdot k}$ for $0 \leq k < n_0$ and $W_{1,\ell} := \tilde{X}_\ell$ for $0 \leq \ell < n_1$.

- **Split**(f) $\rightarrow (f_0^*, \dots, f_{m_0-1}^*)$: Given a polynomial

$$f = \sum_{i=0}^{m_0-1} \sum_{j=0}^{m_1-1} \sum_{k=0}^{n_0-1} \sum_{\ell=0}^{n_1-1} f_{i,j,k,\ell} \cdot V_{0,i} V_{1,j} W_{0,k} W_{1,\ell} \in \mathcal{F},$$

output augmented subpolynomials $f_i^* \in \mathcal{F}^*$ for $0 \leq i < m_0$ as

$$f_i^* := \sum_{j=0}^{m_1-1} \sum_{k=0}^{n_0-1} \sum_{\ell=0}^{n_1-1} f_{i,j,k,\ell} \cdot V_{1,j} W_{0,k} W_{1,\ell} + X^* \cdot \sum_{k=0}^{n_0-1} \sum_{\ell=0}^{n_1-1} g_{i,k,\ell} \cdot W_{0,k} W_{1,\ell},$$

where $g_{i,k,\ell} \stackrel{\$}{\leftarrow} \mathbb{F}$ for $0 \leq i < m_0$, $0 \leq k < n_0$, and $0 \leq \ell < n_1$.

- **EcdToMat**(f^*) $\rightarrow \hat{\mathbf{F}}^*$: Given an augmented subpolynomial

$$f^* = \sum_{j=0}^{m_1-1} \sum_{k=0}^{n_0-1} \sum_{\ell=0}^{n_1-1} f_{j,k,\ell} \cdot V_{1,j} W_{0,k} W_{1,\ell} + X^* \cdot \sum_{k=0}^{n_0-1} \sum_{\ell=0}^{n_1-1} g_{k,\ell} \cdot W_{0,k} W_{1,\ell} \in \mathcal{F}^*,$$

return a matrix $\hat{\mathbf{F}}^* \in R^{(m_1+1) \times n_0}$ generated as follows.

- 1) Define $\mathbf{f}_{j,k} := (f_{j,k,0}, \dots, f_{j,k,n_1-1}) \in \mathbb{F}^{n_1}$ and $\hat{f}_{j,k} := \text{Ecd}(\mathbf{f}_{j,k})$ for $0 \leq j < m_1$ and $0 \leq k < n_0$.
- 2) Define $\mathbf{g}_k := (g_{k,0}, \dots, g_{k,n_1-1}) \in \mathbb{F}^{n_1}$ and $\hat{g}_k := \text{Ecd}(\mathbf{g}_k)$ for $0 \leq k < n_0$.
- 3) Output $\hat{\mathbf{F}}^* := [\hat{\mathbf{f}}_0^* \mid \dots \mid \hat{\mathbf{f}}_{m_1-1}^*] \in R^{(m_1+1) \times n_0}$, where $\hat{\mathbf{f}}_k^* := (\hat{f}_{0,k}, \dots, \hat{f}_{m_1-1,k}, \hat{g}_k) \in R^{m_1+1}$ for $0 \leq k < n_0$.

- **DcdToPoly**($\hat{\mathbf{F}}^*, \mathbf{c}, \alpha$) $\rightarrow f^*$: Given a matrix $\hat{\mathbf{F}}^* = [\hat{\mathbf{f}}_0^* \mid \dots \mid \hat{\mathbf{f}}_{m_1-1}^*] \in R^{(m_1+1) \times n_0}$, $\mathbf{c} \in (R_{X^{\gamma-b}}^\times)^{n_0}$, and $\alpha \in R_{X^{\gamma-b}}^\times$, return an augmented subpolynomial $f^* \in \mathcal{F}^*$ computed as follows.

- 1) Let $\mathbf{c} = (c_0, \dots, c_{n_0-1})$. For $0 \leq j < m_1$ and $0 \leq k < n_0$, define

$$\mathbf{f}_{j,k} := \text{Dcd}(\alpha^{-1} c_k^{-1} \hat{f}_{j,k}), \quad \mathbf{g}_k := \text{Dcd}(\alpha^{-1} c_k^{-1} \hat{g}_k),$$

where $\hat{\mathbf{f}}_k^* = (\hat{f}_{0,k}, \dots, \hat{f}_{m_1-1,k}, \hat{g}_k) \in R^{m_1+1}$.

- 2) Return

$$f^* := [\mathbf{V}_1^\top X^*] \begin{bmatrix} \mathbf{f}_{0,0}^\top & \dots & \mathbf{f}_{0,n_0-1}^\top \\ \vdots & & \vdots \\ \mathbf{f}_{m_1-1,0}^\top & \dots & \mathbf{f}_{m_1-1,n_0-1}^\top \\ \mathbf{g}_0^\top & \dots & \mathbf{g}_{n_0-1}^\top \end{bmatrix} (\mathbf{W}_0 \otimes \mathbf{W}_1).$$

- **Com**(pp, f^*) $\rightarrow (\text{com}, \text{op}^*)$: Given an augmented subpolynomial $f^* \in \mathcal{F}^*$, return a commitment $\text{com}^* = \hat{\mathbf{u}}$ and an opening $\text{op}^* = (\hat{\mathbf{F}}^*, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{c}, 1)$ computed as follows.

- 1) Compute $\hat{\mathbf{F}}^* = [\hat{\mathbf{f}}_0^* \mid \dots \mid \hat{\mathbf{f}}_{m_1-1}^*] := \text{EcdToMat}(f^*)$.
- 2) Sample $\mathbf{R} = [\mathbf{r}_0 \mid \dots \mid \mathbf{r}_{n_0-1}] \stackrel{\$}{\leftarrow} \chi^{(\mu+\nu) \times n_0}$.
- 3) Compute $[\mathbf{t}_0 \mid \dots \mid \mathbf{t}_{n_0-1}] = \mathbf{A} \hat{\mathbf{F}}^* + \mathbf{B} \mathbf{R} \pmod{q}$ and $\hat{\mathbf{t}}_k = \lfloor \mathbf{t}_k / B_t \rfloor$ for $0 \leq k < n_0$. Let $\hat{\mathbf{t}} := \mathbf{t}_0 \parallel \dots \parallel \mathbf{t}_{n_0-1} \in R^{\mu n_0}$.
- 4) Compute $\mathbf{u} = \mathbf{D} \hat{\mathbf{t}} \pmod{q_0}$ and $\hat{\mathbf{u}} = \lfloor \mathbf{u} / B_u \rfloor \in R^\kappa$.
- **Open** $^{(B, B_o, B_c, B_\alpha)}(\text{pp}, \text{com}^*, f^*, \text{op}^*) \rightarrow 0/1$: Given an augmented subpolynomial $f^* \in \mathcal{F}^*$, a commitment $\text{com}^* = \hat{\mathbf{u}}$, and an opening $\text{op}^* = (\hat{\mathbf{F}}^*, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{c}, \alpha)$, where $\mathbf{c} = (c_0, \dots, c_{n_0-1}) \in R^{n_0}$ and $\alpha \in R$, output 1 if the following conditions hold.
 - 1) Check $\|\alpha\|_1 \leq B_\alpha$ and $\alpha \in R_{X^{\gamma-b}}^\times$.
 - 2) Check $\|c_k\|_1 \leq B_c$ and $c_k \in R_{X^{\gamma-b}}^\times$ for $0 \leq k < n_0$.
 - 3) Define $\mathbf{s} := \mathbf{D} \hat{\mathbf{t}} - B_u \alpha \cdot \hat{\mathbf{u}} \pmod{q_0}$.
 - 4) Check $\|\mathbf{s}\|_2 + \|\hat{\mathbf{t}}\|_2 \leq B_o$.
 - 5) Define $\mathbf{h}_k := \mathbf{A} \hat{\mathbf{f}}_k^* + \mathbf{B} \mathbf{r}_k - B_t c_k \cdot \hat{\mathbf{t}}_k \pmod{q}$ for $0 \leq k < n_0$.

- 6) Check $\|\hat{\mathbf{f}}_k^*\|_2 + \|\mathbf{r}_k\|_2 + \|\mathbf{h}_k\|_2 \leq B$ for $0 \leq k < n_0$.

- 7) Check $f^* = \text{DcdToPoly}(\hat{\mathbf{F}}^*, \mathbf{c}, \alpha)$.

- **Com**(pp, f) $\rightarrow (\text{com}, \text{op})$: Given a polynomial $f \in \mathcal{F}$, return a commitment $\text{com} = (\hat{\mathbf{u}}_i)_{i=0}^{m_0-1}$ and an opening $\text{op} = (\text{op}_i^*)_{i=0}^{m_0-1}$ generated as follows.

- 1) Generate $(f_0^*, \dots, f_{m_0-1}^*) \leftarrow \text{Split}(f)$.
- 2) Generate $(\hat{\mathbf{u}}_i, \text{op}_i^*) \leftarrow \text{Com}^*(\text{pp}, f_i^*)$ for $0 \leq i < m_0$.

- **Open** $^{(B, B_o, B_c, B_\alpha)}(\text{pp}, \text{com}, f, \text{op}) \rightarrow 0/1$: Given a polynomial $f \in \mathcal{F}$, a commitment $\text{com} = (\hat{\mathbf{u}}_i)_{i=0}^{m_0-1}$, and an opening $\text{op} = (\text{op}_i^*)_{i=0}^{m_0-1}$, output 1 if the following conditions hold.

- 1) Parse $\text{op}_i^* = (\hat{\mathbf{F}}_i^*, \mathbf{R}_i, \hat{\mathbf{t}}_i, \mathbf{c}_i, \alpha_i)$ for $0 \leq i < m_0$.
- 2) Define $f_i^* := \text{DcdToPoly}(\hat{\mathbf{F}}_i^*, \mathbf{c}_i, \alpha_i)$ for $0 \leq i < m_0$.
- 3) Check $\text{Open}^{(B, B_o, 1, 1)}(\text{pp}, \hat{\mathbf{u}}_i, f_i^*, \text{op}_i^*) = 1$ for $0 \leq i < m_0$.
- 4) Check $f = \sum_{i=0}^{m_0-1} f_i \cdot V_{0,i}$, where $f_i^* = f_i + X^* \cdot g_i$ for $0 \leq i < m_0$.

- **Eval**: Run the composed interactive protocol $\Pi_{\text{Quad}} \circ \Pi_{\text{Agg}} \circ \Pi_{\text{Split}}$, where the subprotocols are defined in Figs. 5 to 7.

Figure 3: Polynomial Commitment Algorithms of Jindo

Commitment Structure. At a high level, the commitment algorithm **Com** of Jindo consists of two steps, **Split** and **Com** * . First, the algorithm runs the **Split** procedure to split an input polynomial $f \in \mathcal{F}$ into augmented subpolynomials $f_i^* = f_i + X^* \cdot g_i \in \mathcal{F}^*$ for $0 \leq i < m_0$. Then, each augmented subpolynomial f_i^* is committed as $\hat{\mathbf{u}}_i$ through the algorithm **Com** * , which yields the outer commitment structure described in Section 2.1. We also note that our outer commitment differs from the construction of Greyhound to improve commitment generation speed. In Greyhound, the digit decomposition of the Ajtai commitments $\mathbf{t}_k$ is committed to reduce the size of q_o , since each entry of $\mathbf{t}_k$ is bounded by q . However, this increases the column dimension of $\mathbf{D}$ and thus degrades prover performance. In contrast, our approach is adapted from the signature compression technique of [BG14], based on the observation that the binding property is preserved when rounding errors are explicitly bounded so that they still yield a short MSIS solution. This reduces the computational overhead of outer commitment generation, since each entry of $\hat{\mathbf{t}}_k$ is now bounded by roughly q/B_t , which permits a smaller q_o while the column dimension of $\mathbf{D}$ remains μn_0 .

Relaxed Opening. For the opening algorithm **Open**, the verifier iteratively runs the subroutine **Open** * for each outer commitment $\hat{\mathbf{u}}_i$ with $0 \leq i < m_0$, and then aggregates the results to determine the validity of the provided opening op . We note that **Open** supports a relaxed opening framework, which is a common technique in lattice-based commitments [NS24, LNP22, BDL⁺18] for handling soundness slack. More precisely, it accepts openings that may differ from the original opening generated during the commitment algorithm **Com** by including relaxation factors $(\mathbf{c}_i, \alpha_i)$ for each i . Although the opening values may differ, the binding property is preserved by ensuring that a commitment com cannot be opened to two distinct polynomials $f \neq f'$. This resolves the gap arising from the extracted witness in relaxed argument of knowledge protocols.

Security Analysis. We now provide a security analysis of

the commitment and opening algorithms of Jindo, focusing on their binding and hiding properties. The binding property relies on the hardness of the MSIS problem, where the norm bound for a short solution is affected by the parameters B , B_o , B_c , and B_α in the opening algorithm Open . At a high level, the binding property for each outer commitment $\hat{\mathbf{u}}_i$ is affected by B_o and B_α , which correspond to the norm bound for openings of $\hat{\mathbf{u}}_i$ and the relaxation factor α_i , respectively. For the inner commitments $\hat{\mathbf{t}}_k$, the binding property is mainly affected by B , which bounds openings of $\hat{\mathbf{t}}_k$, as well as by the norm bounds B_c and B_α for the relaxation factors $\mathbf{c}_i$ and α_i . The hiding property follows from the indistinguishability of the inner commitments $\hat{\mathbf{t}}_k$ from random samples, which is based on the hardness of the MLWE problem. The detailed analysis is as follows.

Theorem 1 (Binding). *The commitment schemes $(\text{Com}, \text{Open})$ and $(\text{Com}^*, \text{Open}^*)$ defined in Fig. 3 satisfy binding if $\text{MSIS}_{q_o, \kappa, 2B_\alpha B_o}$ and $\text{MSIS}_{q, \mu, 2B_\alpha B_c B}$ are computationally hard.*

Theorem 2 (Hiding). *The commitment schemes $(\text{Com}, \text{Open})$ and $(\text{Com}^*, \text{Open}^*)$ defined in Fig. 3 satisfy hiding if $\text{MLWE}_{q, \nu, \chi}$ is computationally hard.*

4.3. Evaluation Protocol

To formally describe our evaluation protocol, we first define two relations, one for the polynomial evaluation relation that we aim to prove and another for the quadratic form evaluation relation to which it is reduced.

- $R_{\text{Poly}}^{(B, B_o, B_c, B_\alpha)}$

$$:= \left\{ (x, w) \mid \text{Open}^{(B, B_o, B_c, B_\alpha)}(\text{pp}, \text{com}, f, \text{op}) = 1 \right\}, \text{ where}$$

$$x = (\text{com}, x, y) \text{ and } w = (f, \text{op}).$$
- $R_{\text{Quad}}^{(B, B_o, B_c, B_\alpha)}$

$$:= \left\{ (x^*, w^*) \mid \hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}^* ((\alpha \mathbf{c})^{-1} \odot \hat{\mathbf{w}}_0) = \hat{y}^* \pmod{X^\gamma - b} \wedge \right. \\ \left. \text{Open}^{*(B, B_o, B_c, B_\alpha)}(\text{pp}, \text{com}^*, f^*, \text{op}^*) = 1 \right\},$$

where $x^* = (\text{com}^*, x^*, \hat{y}^*)$ and $w^* = (f^*, \text{op}^*)$ with $\text{op}^* = (\hat{\mathbf{F}}^*, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{c}, \alpha)$.

$\hat{\mathbf{v}}_1^* = (\text{Ecd}(v_{1,0}), \dots, \text{Ecd}(v_{1,m_1-1}), \text{Ecd}(x^*)) \in R^{m_1+1}$
 $\hat{\mathbf{w}}_0 = (\text{Ecd}(w_{0,0}), \dots, \text{Ecd}(w_{0,n_0-1})) \in R^{n_0}$

Figure 4: Relations for the Jindo PCS

We note that the quadratic form relation checked in R_{Quad} is defined as

$$\hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}^* ((\alpha \mathbf{c})^{-1} \odot \hat{\mathbf{w}}_0) = \hat{y}^* \pmod{X^\gamma - b}.$$

The above relation incorporates the relaxation factors $\mathbf{c}$ and α in addition to the matrix $\hat{\mathbf{F}}^*$ for the quadratic form, as the witness extracted from the evaluation protocol only guarantees a relaxed opening. Consequently, correction by the relaxation factors is required to preserve correct evaluation to the target value $\hat{y}^*$. Once the target relation R_{Poly} is defined, we can check that our commitment algorithm satisfies correctness as follows.

Theorem 3 (Correctness). *Let $B_\chi > 0$ be such that $\|r\|_1 \leq B_\chi$ for every $r \stackrel{\$}{\leftarrow} \chi$. The polynomial commitment scheme defined in Fig. 3 satisfies correctness for the relation $R_{\text{Poly}}^{(B, B_o, 1, 1)}$ with the following norm bounds.*

- $B = b\sqrt{(m_1+1)d} + B_\chi\sqrt{(\mu+\nu)d} + B_t\sqrt{\mu d}$.
- $B_o = (q/B_t)\sqrt{\mu n_0 d} + B_u\sqrt{\kappa d}$.

The theorem above provides a base relation $R_{\text{Poly}}^{(B, B_o, 1, 1)}$ for an honest prover, which is satisfied when a commitment com and an opening op are honestly generated by the algorithm Com . We also note that both relaxation factors are set to 1, since this relation captures the honestly generated opening rather than an extracted one.

We now describe the evaluation protocol of Jindo, which consists of three subprotocols, namely splitting, aggregation, and quadratic form evaluation, as outlined in Section 2.3.

Splitting Protocol. The main goal of the splitting protocol Π_{Split} defined in Fig. 5 is to reduce a polynomial relation R_{Poly} to multiple quadratic form relations R_{Quad} using the augmented subpolynomial structure, while maintaining simulatability for evaluation hiding. The detailed analysis is as follows.

Π_{Split}

Instance
 $x = ((\hat{\mathbf{u}}_i)_{i=0}^{m_0-1}, x, y)$

Witness
 $w = (f, (\hat{\mathbf{F}}_i^*, \mathbf{R}_i, \hat{\mathbf{t}}_i, 1, 1)_{i=0}^{m_0-1})$

Split.P(pp, x, w)
for $0 \leq i < m_0$:
1 : $f_i + X^* \cdot g_i$
2 : $\quad \quad \quad := \text{DcdToPoly}(\hat{\mathbf{F}}_i^*, 1, 1)$
3 : $y_g := \sum_{i=0}^{m_0-1} g_i(x) \cdot v_{0,i} \xrightarrow{y_g}$
4 : $\quad \quad \quad \xleftarrow{x^*} x^* \stackrel{\$}{\leftarrow} \mathbb{F}^\times$
5 : $x^* := (x, x^*) \quad \quad \quad x^* := (x, x^*)$
6 : $\hat{\mathbf{v}}_1^* := (\hat{\mathbf{v}}_1, \text{Ecd}(x^*))$
7 : **for** $0 \leq i < m_0$:
8 : $f_i^* := f_i + X^* \cdot g_i$
9 : $\hat{y}_i^* := \hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}_i^* \hat{\mathbf{w}}_0 \xrightarrow{(\hat{y}_i^*)_{i=0}^{m_0-1}}$
10 : $y + x^* \cdot y_g$
11 : $\quad \quad \quad \stackrel{?}{=} \sum_{i=0}^{m_0-1} \langle \text{Dcd}(\hat{y}_i^*), \mathbf{w}_1 \rangle \cdot v_{0,i}$

Split.V(pp, x)

Reduced Instance
 $x_i^* = (\hat{\mathbf{u}}_i, x^*, \hat{y}_i^*)$ for $0 \leq i < m_0$

Reduced Witness
 $w_i^* = (f_i^*, (\hat{\mathbf{F}}_i^*, \mathbf{R}_i, \hat{\mathbf{t}}_i, 1, 1))$ for $0 \leq i < m_0$

Figure 5: Splitting Protocol of Jindo

Theorem 4 (Splitting Protocol). *Let $\mathbb{F}$ be a finite field such that $1/|\mathbb{F}^\times| = \text{negl}(\lambda)$. The interactive pro-*

tolocol Π_{Split} defined in Fig. 5 is a relaxed reduction of knowledge from $(R_{\text{Poly}}^{(B, B_o, 1, 1)}, R_{\text{Poly}}^{(B', B'_o, B'_c, B'_\alpha)})$ to $(\prod_{i=0}^{m_0-1} R_{\text{Quad}}^{(B, B_o, 1, 1)}, \prod_{i=0}^{m_0-1} R_{\text{Quad}}^{(B', B'_o, B'_c, B'_\alpha)})$.

Aggregation Protocol. Once the splitting protocol reduces a polynomial relation R_{Poly} to multiple quadratic form relations in R_{Quad} , the role of the aggregation protocol defined in Fig. 6 is to further reduce them to a single quadratic form relation in R_{Quad} . Another crucial role of the aggregation protocol is to ensure simulatability of $(\hat{\mathbf{F}}^*, \mathbf{R})$ in the reduced witness, which is needed for simulatability in the subsequent single quadratic form evaluation protocol.

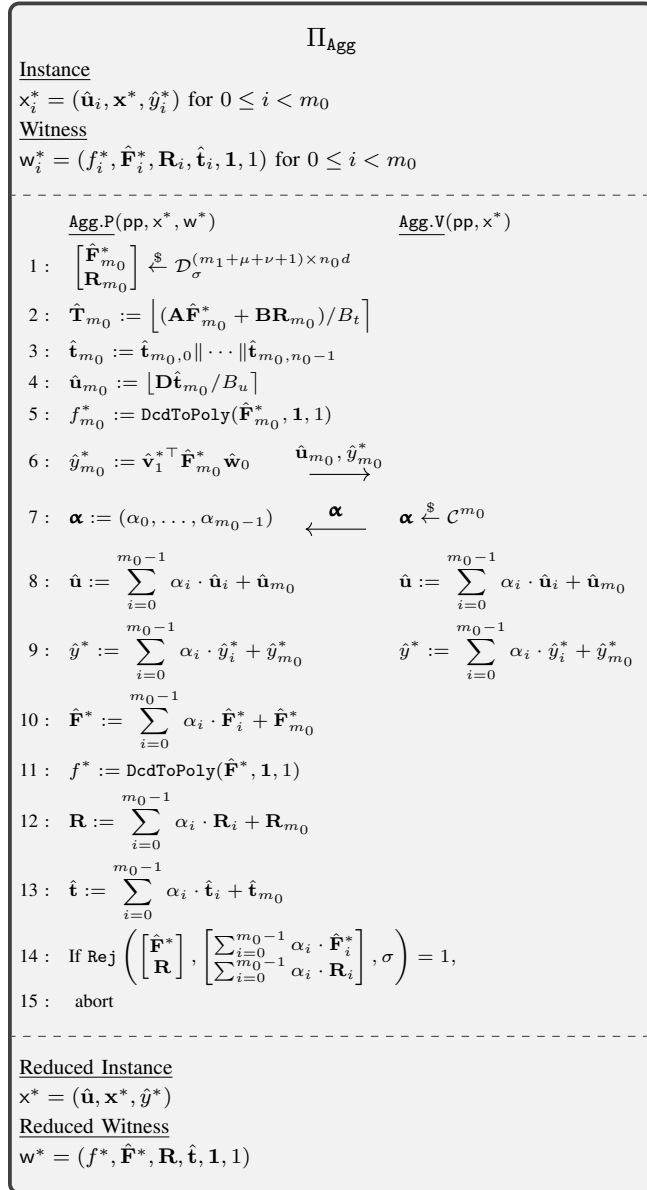


Figure 6: Aggregation Protocol of Jindo

Theorem 5 (Aggregation Protocol). *Let $\mathcal{C} \subseteq R$ be a η -well-spread set for R_p such that $\eta = \text{negl}(\lambda)$, and let $B_C > 0$ be*

such that $\|c\|_1 \leq B_C$ for all $c \in \mathcal{C}$. The interactive protocol Π_{Agg} defined in Fig. 6 is a relaxed reduction of knowledge from $(\prod_{i=0}^{m_0-1} R_{\text{Quad}}^{(B, B_o, 1, 1)}, \prod_{i=0}^{m_0-1} R_{\text{Quad}}^{(B', B'_o, B'_c, B'_\alpha)})$ to $(R_{\text{Quad}}^{(B, B_o, 1, 1)}, R_{\text{Quad}}^{(B', B'_o, B'_c, 1)})$ with acceptance rate at least $(1 - 2^{-128})/M$, where the norm bounds are defined as follows.

- $\sigma = \frac{14}{\ln M} m_0 \sqrt{n_0} B_C B$.
- $B = m_0 B_C B + \sigma (\sqrt{2(m_1 + 1)d} + \sqrt{2(\mu + \nu)d}) + B_t \sqrt{\mu d}$.
- $B_o = m_0 B_C B_o + B_u \sqrt{\kappa d} + (q/B_t) \sqrt{\mu n_0 d}$.
- $B' = 2B'_c B'$, $B'_o = 2B'_o$, $B'_c = (B'_c)^2$, and $B'_\alpha = 2B_C$.

Quadratic Form Evaluation Protocol. As the final step, we present the single quadratic form evaluation protocol in Fig. 7, which is based on the construction of Greyhound. The key difference from Greyhound is that we allow a challenge set whose invertibility condition depends only on the message space $R_{X^\gamma - b}$ rather than the commitment space R_q . To ensure invertibility over $R_{X^\gamma - b} = \mathbb{Z}_p[X]/(X^\gamma - b)$, we use well-spreadness over $R_p = \mathbb{Z}_p[X]/(X^d + 1)$, since $X^\gamma - b$ divides $X^d + 1$ modulo p . We also note that the verification of the final response is modified to incorporate the rounding compression used in the inner and outer commitments.

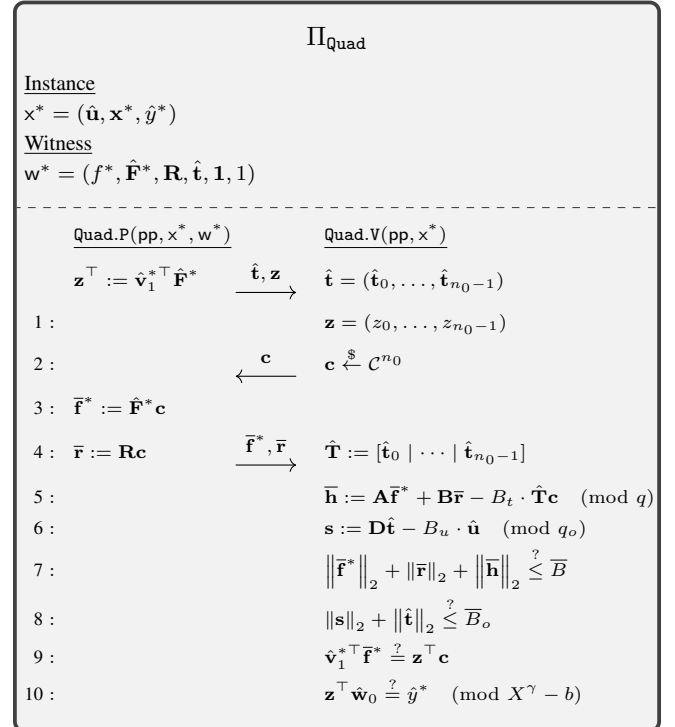


Figure 7: Quadratic Form Evaluation Protocol of Jindo

Theorem 6 (Quadratic Form Evaluation Protocol). *Let $\mathcal{C} \subseteq R$ be an η -well-spread set for R_p such that $\eta = \text{negl}(\lambda)$, and let $B_C > 0$ be such that $\|c\|_1 \leq B_C$ for all $c \in \mathcal{C}$. Then, the interactive protocol $(\text{Quad.P}, \text{Quad.V})$ defined in Fig. 7 is a relaxed interactive argument of knowledge for $(R_{\text{Quad}}^{(B, B_o, 1, 1)}, R_{\text{Quad}}^{(B', B'_o, B'_c, 1)})$ with failure probability 0, where the norm bounds are defined as follows.*

- $\bar{B} = n_0 B_C B$ and $\bar{B}_o = B_o$.
- $B' = 2\bar{B} = 2n_0 B_C B$, $B'_o = \bar{B}_o = B_o$, and $B'_c = 2B_C$.

Polynomial Evaluation Protocol. Finally, composing all these protocols yields a polynomial evaluation protocol that achieves evaluation hiding and $\tilde{O}(N^{1/3})$ complexity in both communication and verification. The detailed analysis is as follows.

Corollary 1 (Polynomial Evaluation Protocol). *The composed protocol $\Pi_{\text{Eval}} := \Pi_{\text{Quad}} \circ \Pi_{\text{Agg}} \circ \Pi_{\text{Split}}$ is a relaxed argument of knowledge for $(R_{\text{Poly}}^{(B, B_o, 1, 1)}, R_{\text{Poly}}^{(B', B'_o, B'_c, B'_\alpha)})$ with acceptance rate at least $(1 - 2^{-128})/M$, when the following norm bounds hold.*

- $B' = (8 + 104\sqrt{2(m_1 + 1)n_0 d} + 104\sqrt{2(\mu + \nu)n_0 d}) m_0 n_0 B_C^3 B + 8n_0 B_t B_C^2 \sqrt{\mu d}$.
- $B'_o = 2m_0 B_C B_o + 2B_u \sqrt{\kappa d} + (2q/B_t) \sqrt{\mu n_0 d}$.
- $B'_c = 4B_C^2$ and $B'_\alpha = 2B_C$.

Theorem 7 (Evaluation Hiding). *Let $\Pi_{\text{Eval}} := \Pi_{\text{Quad}} \circ \Pi_{\text{Agg}} \circ \Pi_{\text{Split}}$ be the evaluation protocol for the polynomial commitment scheme defined in Fig. 3. Then, it achieves evaluation hiding with advantage at most $\text{negl}(\lambda) + 2^{-128}$ if $\text{MLWE}_{q, \nu, \chi}$ is computationally hard and $\sigma = \frac{14}{\ln M} m_0 \sqrt{n_0 B_C B}$.*

Theorem 8 (Complexity). *The evaluation protocol $\Pi_{\text{Eval}} = \Pi_{\text{Quad}} \circ \Pi_{\text{Agg}} \circ \Pi_{\text{Split}}$ achieves the following complexity.*

- Commitment complexity: $O(N \log^2(N))$
- Evaluation complexity: $O(N)$
- Verification complexity: $O(N^{1/3} \cdot \log^{10/3}(N))$
- Communication complexity: $O(N^{1/3} \cdot \log^{4/3}(N))$.

5. Implementation

We implemented our PCS, Jindo, to demonstrate its practical efficiency. We begin by presenting the parameter settings and optimization techniques used in our implementation. Next, we benchmark single polynomial evaluation to demonstrate the concrete performance of our PCS in comparison with other PCS constructions. Finally, we assess the performance of Jindo within PIOP compilations to showcase its practical advantages in constructing PIOP-based ZKAoKs for client-side proving. All benchmarks were conducted in a single-threaded setting on a laptop equipped with an Intel i5-11300H CPU running at 3.10 GHz and 16 GB of RAM. Our source code is available at <https://github.com/sp301415/ringo-snark>.

5.1. Parameter Settings and Optimizations

We present the parameter settings and optimization techniques used in our implementation. We also provide a script in our source code that automatically selects the best performing parameters for a given coefficient space $\mathbb{F}$ and polynomial size N .

MSIS and MLWE. For the hardness of the MSIS problem, we use Eq. (2) with root Hermite factor $\delta \approx 1.005$, following prior works [HSS24, ENS20, LNS20]. For the hardness

of the MLWE problem, we use the estimation algorithm of [APS15]. Based on these analyses, all MSIS and MLWE parameters are configured to provide 128-bit security.

Encoding. For the encoding method, we implement the coefficient encoding φ_{Coeff} from Lemma 5. For the parameter settings, we found that setting b around 32-bit yields favorable performance in terms of both proof size and proof generation speed.

Challenge Set. For the challenge set $\mathcal{C}$, we use the set S_ω and analyze its spreadness over R_p based on Lemma 3. However, directly computing Eq. (1) in Lemma 3 requires $O(p)$ time, which is infeasible when p is exponential in the security parameter. We therefore use the heuristic analysis presented in [ESZ22], following prior works [ESZ22, AAB⁺24, GLLO26, LNPS21] that also rely on the spreadness of the set S_ω . Based on this analysis, we use the minimum value of w that provides $\eta \approx 2^{-128}$ for the given coefficient modulus p and ring degree d .

Rejection Sampling with Rounded Gaussian. To reduce sampling overhead, we use the rounded Gaussian distribution instead of the discrete Gaussian distribution, which is much faster to sample. We specifically observe that the statistical distance between the rounded Gaussian distribution and the discrete Gaussian distribution is bounded by $O(\sigma^{-2})$, where σ is the standard deviation. In our parameter setting, $\sigma \approx 2^{50}$, which yields statistical distance about 2^{-100} between the discrete Gaussian distribution and the rounded Gaussian distribution. Therefore, in our implementation, we use the rounded Gaussian distribution to improve proving performance, while lowering the statistical security of evaluation hiding to 100-bit, which still provides strong practical security. We present a more detailed analysis in Theorem 9.

Optimization for Polynomial Arithmetic. We use SIMD instructions, such as AVX512-IFMA, to accelerate polynomial arithmetic over $R_q = \mathbb{Z}_q[X]/(X^d + 1)$ and $R_{q_o} = \mathbb{Z}_{q_o}[X]/(X^d + 1)$, following [NS24, BKS⁺21]. Beyond SIMD acceleration, we choose NTT-friendly moduli for both q and q_o , namely moduli satisfying $2d \mid q - 1$ and $2d \mid q_o - 1$, which enable fast polynomial multiplication via the number theoretic transform (NTT). We note that some lattice-based proof systems, such as Lazer [LSS24], Labrador [BS23], and Greyhound [NS24], do not support such moduli because of constraints from knowledge extraction, which introduces additional overhead. In contrast, our PCS imposes no such constraints on the commitment moduli thanks to our new extraction method.

5.2. Benchmark for PCS

In Table 1, we provide benchmark results for Jindo together with the performance of other PCSs. At a high level, these schemes can be grouped by their underlying cryptographic assumptions. Hyrax [WTS⁺18] is based on the discrete logarithm problem, while Ligerio [AHIV17], WHIR [ACFY25], DeepFold [GLH⁺25], and PIP-FRI [LZC⁺26] are constructed from random oracles and

Prover (ms)					Verifier (ms)					Proof Size (KB)				
$\log N$	14	16	18	20	$\log N$	14	16	18	20	$\log N$	14	16	18	20
Hyrax	346	1159	4016	13681	Hyrax	12	21	39	73	Hyrax	12	25	49	98
Ligero	42	106	315	1089	Ligero	37	61	110	207	Ligero	370	606	1068	1980
WHIR	35	115	427	1813	WHIR	0.9	1.1	1.2	1.4	WHIR	198	232	256	291
zk-DeepFold	81	358	1480	6619	zk-DeepFold	4	6	9	12	zk-DeepFold	329	482	675	905
zk-PIP-FRI	37	167	712	2927	zk-PIP-FRI	8	13	18	24	zk-PIP-FRI	879	1128	1344	1593
Greyhound	19	49	159	695	Greyhound	8	17	42	96	Greyhound	37	40	40	46
zk-CELPC	138	358	1087	3714	zk-CELPC	29	52	98	168	zk-CELPC	1474	2526	4646	8922
zk-Jindo	14	32	83	273	zk-Jindo	5	7	10	14	zk-Jindo	315	474	745	1207

TABLE 1: Benchmark Results for Single Polynomial Evaluation.

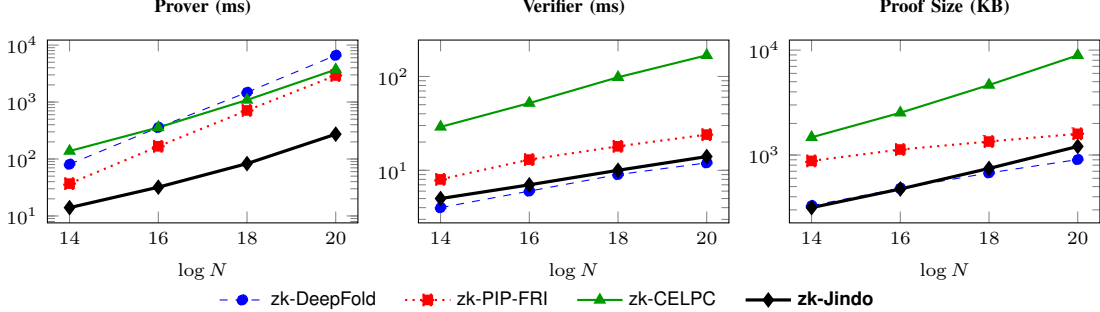


Figure 8: Benchmark Results for Evaluation Hiding Post-quantum PCS

error correcting codes, in particular Reed Solomon codes. Finally, Greyhound [NS24], CELPC [HSS24], and Jindo are based on lattice cryptography. For the benchmarks, we use the reference implementation provided in each paper, except for Hyrax and Ligero, for which we use the implementation available at <https://github.com/arkworks-rs/poly-commit>. Among these PCS implementations, only DeepFold, PIP-FRI, and CELPC, together with our implementation of Jindo, provide evaluation hiding. We therefore add the prefix zk to these PCSs.

For the parameter setting, we use an approximately 256 bit prime field for all PCSs except for Greyhound, whose implementation supports only a 32 bit prime field. Thus, we measure the performance of Greyhound with a polynomial degree of $8N$ to obtain a fair comparison, since this equalizes the input witness size as $8 = 256/32$. For code based PCSs, we use a code rate of $\rho = 1/2$ because this yields the fastest proving performance. The bound for list decodability is set to the Johnson bound $\approx 1 - \sqrt{\rho}$ rather than the conjectured optimal capacity bound $\approx 1 - \rho$.

When comparing Jindo with other evaluation hiding PCSs, Fig. 8 shows that Jindo achieves about an order of magnitude faster proving performance than the other schemes, while providing verification time and proof size comparable to those of DeepFold, which achieves the best results for these metrics. In particular, Jindo improves upon CELPC by about an order of magnitude across all metrics, making it a superior alternative for applications that currently use CELPC [HLSS25, HMSS25a, HMSS25b].

Even when Jindo is compared with PCSs without evaluation hiding, it still achieves significantly faster proving performance. For example, when $\log N = 20$, Jindo is about 50 times faster than Hyrax, 4 times faster than Ligero, 6.5 times

faster than WHIR, and 2.5 times faster than Greyhound. These performance gap will be further widened if evaluation hiding overhead are added to each of them. Finally, we note that the verification cost of Jindo is significantly lower than those of other lattice based PCSs, namely Greyhound and CELPC, thanks to the complexity improvement from square root to cube root.

5.3. Benchmark for PIOP

To demonstrate the effectiveness of Jindo in ZKAoK constructions, we measure its performance in PIOP compilation scenarios. Specifically, we measure the performance of ZKAoKs for the validity of RLWE samples based on the PIOP construction of [HLSS25], compiled with Jindo and CELPC, respectively. Beyond the PIOP based approach, we also provide benchmark results for Lazer [LSS24], which supports ZKAoKs for RLWE samples based on the ABDLOP commitment system [LNP22]. For the target statement, we prove the validity of an RLWE sample $(a, b) \in R_q = \mathbb{Z}_q[X]/(X^d + 1)$ by showing that it satisfies the relation $b = as + e$ for some $s, e \in R_q$ with ternary coefficients. We measure performance while varying the parameters $\log q$ and d to demonstrate the applicability of Jindo across various parameter regimes. For Lazer, we do not report performance for $\log q \approx 440$, since its reference implementation does not support this parameter regime. In contrast, Jindo supports this parameter regime thanks to the flexibility of the CELPC style encoding. The benchmark results are presented in Table 2.

Compared with CELPC, Jindo improves the performance of ZKAoKs by about an order of magnitude across all measured metrics. We attribute this improvement to the performance gains of the Jindo PCS over CELPC, which

also achieves about an order of magnitude improvement across all metrics, as noted above. Compared with **Lazer**, **Jindo** achieves about an order of magnitude improvement in proof generation speed and about two orders of magnitude improvement in verification speed, while producing proofs that are about four times larger. We note that the cost of the larger proof size is partly offset when the size of the RLWE sample (a, b) is included in the total communication cost. For example, for the parameter setting $(220, 2^{12})$, the overall communication cost of **Jindo** is only about twice that of **Lazer**.

(log q, d)		$(110, 2^{12})$	$(220, 2^{13})$	$(440, 2^{14})$
Prover (ms)	Lazer	1168	5503	-
	CELPC	1097	3119	9214
	Jindo	75	229	876
Verify (ms)	Lazer	620	3069	-
	CELPC	193	467	995
	Jindo	13	31	96
Proof (KB)	Lazer	75	142	-
	CELPC	3506	5204	9103
	Jindo	367	576	901

TABLE 2: Benchmark Results for Proof of RLWE Samples

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Appendix A. Rounded Gaussian Distribution

Definition 11 (Smoothing Parameter [MR07]). *For an n -dimensional lattice Λ and positive real $\varepsilon > 0$, the smoothing parameter $\eta_\varepsilon(\Lambda)$ is the smallest s such that $\rho_{1/s}(\Lambda^* \setminus \{0\}) \leq \varepsilon$, where $\rho_{1/s}(x) := \exp(-\pi s^2 x^2)$.*

Lemma 9 (Lemma 3.3 [MR07]). *The following bound holds for $0 < \varepsilon < 1$.*

$$\eta_\varepsilon(\mathbb{Z}) \leq \sqrt{\frac{\ln(2n(1 + 1/\varepsilon))}{\pi}}$$

Lemma 10 (Lemma 2.6 [GMPW20]). *Let $s \geq \eta_\varepsilon(\mathbb{Z})$, then $\rho_s(c + \mathbb{Z}) \in [(1 - \varepsilon)s, (1 + \varepsilon)s]$ for any $c \in \mathbb{R}$.*

Definition 12 (Rounded Gaussian). *The rounded Gaussian distribution on $\mathbb{Z}$ with parameter $\sigma > 0$ is defined as $\mathcal{R}_\sigma(x) := \int_{x-1/2}^{x+1/2} \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{t^2}{2\sigma^2}) dt$.*

Lemma 11 (Lemma C.2 [HLS18]).

- 1) *For any $k\sigma > 1/4(\sigma + 1)$, $\sigma \geq 1$, $\Pr[|z| > k\sigma \mid z \xleftarrow{\$} \mathcal{R}_\sigma] \leq 2e^{-(k-1/2)^2/2}$.*
- 2) *For any $k > 1$, $\Pr[\|z\|_2 > k\sigma\sqrt{m} \mid z \xleftarrow{\$} \mathcal{R}_\sigma^m] < 2k^m e^{m(1-k^2)/2}$.*

Theorem 9. *Let $0 < \varepsilon < 1/2$ and $\sigma \geq \frac{1}{\sqrt{2\pi}} \cdot \eta_\varepsilon(\mathbb{Z}) + 1$. Let $k > 1$ such that $\Pr[|z| > k\sigma \mid z \xleftarrow{\$} \mathcal{R}_\sigma] \leq \varepsilon$ and $\Pr[|z| > k\sigma \mid z \xleftarrow{\$} \mathcal{D}_\sigma] \leq \varepsilon$. Then, the following holds*

$$\Delta(\mathcal{R}_\sigma, \mathcal{D}_\sigma) < \frac{k(k+2)(1 + \varepsilon + 1/\sqrt{2\pi}\sigma^2)}{\sigma^2} + \varepsilon(3 + \varepsilon)$$

where Δ is the statistical distance.

Proof. Let $f(x) := \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{x^2}{2\sigma^2})$, then we have

$$\Delta(\mathcal{R}_\sigma, \mathcal{D}_\sigma) = \sum_{x \in \mathbb{Z}} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} f(t) dt - \frac{f(x)}{f(\mathbb{Z})} \right|.$$

By Lemma 10, $f(\mathbb{Z}) = \frac{1}{\sqrt{2\pi}\sigma} \rho_{\sqrt{2\pi}\sigma}(\mathbb{Z}) \in [1 - \varepsilon, 1 + \varepsilon]$ since $\sigma \geq \frac{1}{\sqrt{2\pi}} \cdot \eta_\varepsilon(\mathbb{Z})$. Thus, $|1 - 1/f(\mathbb{Z})| \leq \frac{\varepsilon}{1 - \varepsilon} < 2\varepsilon$, which yields

$$\begin{aligned} & \sum_{x \in \mathbb{Z}} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} f(t) dt - \frac{f(x)}{f(\mathbb{Z})} \right| \\ &= \sum_{x \in \mathbb{Z}} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} f(t) dt - f(x) + (1 - 1/f(\mathbb{Z}))f(x) \right| \end{aligned}$$

$$\begin{aligned}
&\leq \sum_{x \in \mathbb{Z}} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} f(t) dt - f(x) \right| + \varepsilon \cdot f(\mathbb{Z}) \\
&< \sum_{x \in \mathbb{Z}} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} f(t) dt - f(x) \right| + \varepsilon(1 + \varepsilon).
\end{aligned}$$

Next, we consider the tail cutoff. By the definition of k , we have the following.

$$\begin{aligned}
&\sum_{x \in \mathbb{Z}} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} f(t) dt - f(x) \right| \\
&\leq \sum_{-k\sigma \leq x \leq k\sigma} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} f(t) dt - f(x) \right| + 2\varepsilon
\end{aligned}$$

Finally, we consider the upper bound of $\int_{x-1/2}^{x+1/2} f(t) dt - f(x) = \int_{x-1/2}^{x+1/2} (f(t) - f(x)) dt$ within the range $[-k\sigma, k\sigma]$. By Taylor's remainder theorem, we have

$$f(t) - f(x) = (t - x)f'(x) + \frac{(t - x)^2}{2} f''(c(t))$$

where $c(t)$ lies between t and x . Since $f''(x) = \frac{x^2 - \sigma^2}{\sigma^4} f(x)$, $|t - x| < 1/2$, and $|c(t)| \leq (k + 1)\sigma$, we have

$$\begin{aligned}
\left| \frac{(t - x)^2}{2} f''(c(t)) \right| &= \left| \frac{(t - x)^2}{2} \cdot \frac{c(t)^2 - \sigma^2}{\sigma^4} f(c(t)) \right| \\
&< \frac{1}{8} \cdot \frac{k(k + 2)}{\sigma^2} f(c(t)).
\end{aligned}$$

Therefore, we have

$$\begin{aligned}
&\sum_{-k\sigma \leq x \leq k\sigma} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} (f(t) - f(x)) dt \right| \\
&= \sum_{-k\sigma \leq x \leq k\sigma} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} \left((t - x)f'(x) + \frac{(t - x)^2}{2} f''(c(t)) \right) dt \right| \\
&= \sum_{-k\sigma \leq x \leq k\sigma} \frac{1}{2} \left| \int_{x-1/2}^{x+1/2} \frac{(t - x)^2}{2} f''(c(t)) dt \right| \\
&< \frac{k(k + 2)}{16\sigma^2} \cdot \sum_{-k\sigma \leq x \leq k\sigma} \left| \int_{x-1/2}^{x+1/2} f(c(t)) dt \right|.
\end{aligned}$$

We note that $f(c(t)) \leq f(0)$ if $x = 0$, $f(c(t)) \leq f(x - 1/2)$ if $x \geq 1$, and $f(c(t)) \leq f(x + 1/2)$ if $x \leq -1$. Thus, we have

$$\begin{aligned}
&\sum_{-k\sigma \leq x \leq k\sigma} \left| \int_{x-1/2}^{x+1/2} f(c(t)) dt \right| \\
&\leq f(0) + \sum_{-k\sigma \leq x \leq -1} f(x + 1/2) + \sum_{1 \leq x \leq k\sigma} f(x - 1/2) \\
&< f(0) + f(\mathbb{Z} + 1/2) < \frac{1}{\sqrt{2\pi}\sigma} + 1 + \varepsilon
\end{aligned}$$

since $f(\mathbb{Z} + 1/2) \leq 1 + \varepsilon$ by Lemma 10. To sum up, the following bound holds.

$$\Delta(\mathcal{R}_\sigma, \mathcal{D}_\sigma) < \frac{k(k + 2)(1 + \varepsilon + 1/\sqrt{2\pi}\sigma^2)}{\sigma^2} + \varepsilon(3 + \varepsilon)$$

Appendix B. Deferred Proofs

B.1. Proof of Lemma 5

Proof. We have $R_{X^\gamma - b} = \mathbb{Z}_p[X]/(X^\gamma - b)$. Since $\mathbb{Z}_p[X]/(X^\gamma - b)$ is a vector space over $\mathbb{Z}_p$ of dimension γ with basis $\{1, X, \dots, X^{\gamma-1}\}$, it follows that $R_{X^\gamma - b} \cong \mathbb{Z}_p^\gamma$ as a vector space. $\square$

B.2. Proof of Lemma 6

Proof. Since $p \equiv 2\delta + 1 \pmod{4\delta}$, the polynomial $X^\gamma - b$ factors over $\mathbb{Z}_p$ as $X^\gamma - b = \prod_{i=0}^{\gamma\delta/d-1} (X^{d/\delta} - \zeta_i)$, where each $X^{d/\delta} - \zeta_i$ is irreducible over $\mathbb{Z}_p$. By the Chinese remainder theorem, we have

$$\begin{aligned}
R_{X^\gamma - b} &= \mathbb{Z}_p[X]/(X^\gamma - b) = \mathbb{Z}_p[X] / \left(\prod_{i=0}^{\gamma\delta/d-1} (X^{d/\delta} - \zeta_i) \right) \\
&\cong \prod_{i=0}^{\gamma\delta/d-1} \mathbb{Z}_p[X]/(X^{d/\delta} - \zeta_i).
\end{aligned}$$

Since each $X^{d/\delta} - \zeta_i$ is irreducible over $\mathbb{Z}_p$, we have $\mathbb{Z}_p[X]/(X^{d/\delta} - \zeta_i) \cong \text{GF}(p^{d/\delta})$ for every $0 \leq i < \gamma\delta/d$. Therefore, there exists an isomorphism $\varphi_{\text{S1ot}} : \text{GF}(p^{d/\delta})^{\gamma\delta/d} \rightarrow R_{X^\gamma - b}$. $\square$

B.3. Proof of Lemma 8

Proof. If $\varphi = \varphi_{\text{S1ot}}$, the relation holds directly since φ_{S1ot} is a ring isomorphism and $\text{Ecd}(\alpha) = \text{Ecd}(\alpha \cdot \mathbf{1})$. It remains to consider the case where $\varphi = \varphi_{\text{Coeff}}$ and $\mathbb{F} = \mathbb{Z}_p$. In this case, $\text{Ecd}(\alpha) = \alpha \pmod{X^\gamma - b}$. Thus,

$$\begin{aligned}
\text{Ecd}(\alpha) \cdot \text{Ecd}(\mathbf{v}) &= \alpha \cdot \varphi_{\text{Coeff}}(\mathbf{v}) = \varphi_{\text{Coeff}}(\alpha \cdot \mathbf{v}) \\
&= \text{Ecd}(\alpha \cdot \mathbf{v}) \pmod{X^\gamma - b},
\end{aligned}$$

since φ_{Coeff} is a vector space isomorphism. $\square$

B.4. Proof of Theorem 1

Proof. Since $(\text{Com}, \text{Open})$ is obtained by repeating $(\text{Com}^*, \text{Open}^*)$, it suffices to prove the binding property of $(\text{Com}^*, \text{Open}^*)$. Suppose that there exists an adversary A against the binding property of $(\text{Com}^*, \text{Open}^*)$ with non-negligible advantage. Then, it outputs $(\hat{\mathbf{u}}, f^*, f'^*, \text{op}^*, \text{op}'^*)$ such that $f^* \neq f'^*$, $\text{Open}^*(\text{pp}, \hat{\mathbf{u}}, f^*, \text{op}^*) = 1$, and $\text{Open}^*(\text{pp}, \hat{\mathbf{u}}, f'^*, \text{op}'^*) = 1$.

Let $\text{op}^* = (\hat{\mathbf{F}}^*, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{c}, \alpha)$, $\mathbf{s} := \mathbf{D}\hat{\mathbf{t}} - B_u\alpha \cdot \hat{\mathbf{u}}$, $\text{op}'^* = (\hat{\mathbf{F}}'^*, \mathbf{R}', \hat{\mathbf{t}}', \mathbf{c}', \alpha')$, and $\mathbf{s}' := \mathbf{D}\hat{\mathbf{t}}' - B_u\alpha' \cdot \hat{\mathbf{u}}$. Then, we have

$$[\mathbf{D} \mid \mathbf{I}_\kappa] \begin{bmatrix} \alpha'\hat{\mathbf{t}} - \alpha\hat{\mathbf{t}}' \\ \alpha\mathbf{s}' - \alpha'\mathbf{s} \end{bmatrix} = \mathbf{0} \pmod{q_o},$$

$$\|\alpha'\hat{\mathbf{t}} - \alpha\hat{\mathbf{t}}'\|_2 + \|\alpha\mathbf{s}' - \alpha'\mathbf{s}\|_2 \leq 2B_\alpha B_o.$$

$\square$ Thus, the hardness of $\text{MSIS}_{q_o, \kappa, 2B_\alpha B_o}$ implies $\alpha'\hat{\mathbf{t}} = \alpha\hat{\mathbf{t}}'$.

Next, let $\mathbf{h}_k := \mathbf{A}\hat{\mathbf{f}}_k^* + \mathbf{B}\mathbf{r}_k - B_t c_k \cdot \hat{\mathbf{t}}_k$ and $\mathbf{h}'_k := \mathbf{A}\hat{\mathbf{f}}_k'^* + \mathbf{B}\mathbf{r}'_k - B_t c'_k \cdot \hat{\mathbf{t}}'_k$ for $0 \leq k < n_0$. Then, for $0 \leq k < n_0$, we have

$$\alpha' c'_k \mathbf{h}_k - \alpha c_k \mathbf{h}'_k = \mathbf{A}(\alpha' c'_k \hat{\mathbf{f}}_k^* - \alpha c_k \hat{\mathbf{f}}_k'^*) + \mathbf{B}(\alpha' c'_k \mathbf{r}_k - \alpha c_k \mathbf{r}'_k) \\ \implies [\mathbf{A} \mid \mathbf{B} \mid \mathbf{I}_\mu] \begin{pmatrix} \alpha' c'_k \hat{\mathbf{f}}_k^* - \alpha c_k \hat{\mathbf{f}}_k'^* \\ \alpha' c'_k \mathbf{r}_k - \alpha c_k \mathbf{r}'_k \\ -\alpha' c'_k \mathbf{h}_k + \alpha c_k \mathbf{h}'_k \end{pmatrix} = \mathbf{0} \pmod{q}.$$

Moreover, the following norm bound holds for $0 \leq k < n_0$.

$$\|\alpha' c'_k \hat{\mathbf{f}}_k^* - \alpha c_k \hat{\mathbf{f}}_k'^*\|_2 + \|\alpha' c'_k \mathbf{r}_k - \alpha c_k \mathbf{r}'_k\|_2 + \|\alpha' c'_k \mathbf{h}_k - \alpha c_k \mathbf{h}'_k\|_2 \\ \leq 2B_\alpha B_c B.$$

Thus, by the hardness of $\text{MSIS}_{q,\mu,2B_\alpha B_c B}$, we obtain

$$\alpha' c'_k \hat{\mathbf{f}}_k^* = \alpha c_k \hat{\mathbf{f}}_k'^* \iff \alpha^{-1} c_k^{-1} \hat{\mathbf{f}}_k^* = \alpha'^{-1} c'_k{}^{-1} \hat{\mathbf{f}}_k'^* \pmod{X^\gamma - b}$$

for all $0 \leq k < n_0$. This implies

$$f^* = \text{DcdToPoly}(\hat{\mathbf{F}}^*, \mathbf{c}, \alpha) = \text{DcdToPoly}(\hat{\mathbf{F}}'^*, \mathbf{c}', \alpha') = f'^*.$$

This contradicts the assumption that $f^* \neq f'^*$. Hence, no such PPT adversary $\mathbf{A}$ exists against the binding property of $(\text{Com}^*, \text{Open}^*)$. $\square$

B.5. Proof of Theorem 2

Proof. Since $(\text{Com}, \text{Open})$ is obtained by repeating $(\text{Com}^*, \text{Open}^*)$, it suffices to prove the hiding property of $(\text{Com}^*, \text{Open}^*)$. To prove hiding, it suffices to consider the distribution of

$$\mathbf{t}_k = \mathbf{A}\hat{\mathbf{f}}_k^* + \mathbf{B}\mathbf{r}_k \pmod{q}$$

for $0 \leq k < n_0$. Recall that $\mathbf{B} = [\mathbf{B}' \mid \mathbf{I}_\mu]$, where $\mathbf{B}' \xleftarrow{\$} R_q^{\mu \times \nu}$, and $\mathbf{r}_k \xleftarrow{\$} \chi^{\mu + \nu}$. Under the hardness of $\text{MLWE}_{q,\nu,\chi}$, the term $\mathbf{B}\mathbf{r}_k$ is computationally indistinguishable from a uniformly random element of R_q^μ . Therefore, for any fixed message column $\hat{\mathbf{f}}_k^*$, the value $\mathbf{t}_k$ is computationally indistinguishable from a uniformly random element of R_q^μ . This holds for all $0 \leq k < n_0$, and hence the rounded inner commitments $\hat{\mathbf{t}}_k = \lfloor \mathbf{t}_k / B_t \rfloor$ reveal no computational information about $\hat{\mathbf{f}}_k^*$. Since the outer commitment $\hat{\mathbf{u}}$ is computed only from the rounded inner commitments, it also reveals no computational information about the committed augmented matrix $\hat{\mathbf{F}}^*$. Thus, $(\text{Com}^*, \text{Open}^*)$ is hiding, and therefore so is $(\text{Com}, \text{Open})$. $\square$

B.6. Proof of Theorem 3

Proof. Let $(\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f)$, where $\text{com} = (\hat{\mathbf{u}}_i)_{i=0}^{m_0-1}$ and $\text{op} = (\text{op}_i^*)_{i=0}^{m_0-1}$ with $\text{op}_i^* = (\hat{\mathbf{F}}_i^*, \mathbf{R}_i, \hat{\mathbf{t}}_i, 1, 1)$. By construction, the algebraic checks in Open^* hold for each $0 \leq i < m_0$. It remains to verify the norm bounds. For $0 \leq i < m_0$ and $0 \leq k < n_0$, we have $\|\hat{\mathbf{f}}_{i,k}^*\|_\infty \leq \frac{b+1}{2}$ by Lemma 7, $\|\mathbf{r}_{i,k}\|_\infty \leq B_\chi$ by the assumption on χ , and $\|\mathbf{h}_{i,k}\|_\infty \leq B_t$ by the rounding definition. Therefore, $\|\hat{\mathbf{f}}_{i,k}^*\|_2 + \|\mathbf{r}_{i,k}\|_2 + \|\mathbf{h}_{i,k}\|_2 \leq B$. Similarly, $\|\hat{\mathbf{t}}_i\|_\infty \leq q/B_t$ and $\|\mathbf{s}_i\|_\infty \leq B_u$ hold by the

rounding definitions. Hence, $\|\hat{\mathbf{t}}_i\|_2 + \|\mathbf{s}_i\|_2 \leq B_o$ for every $0 \leq i < m_0$. Thus, $\text{Open}^{(B, B_o, 1, 1)}(\text{pp}, \text{com}, f, \text{op}) = 1$, which proves correctness. $\square$

B.7. Proof of Theorem 4

Proof. Completeness and public reducibility follow directly from the protocol, so we only consider knowledge soundness.

Knowledge Soundness. Suppose there exists a deterministic prover $\mathbf{P}^*$ that accepts with non-negligible probability $\varepsilon > 1/|\mathbb{F}^\times|$. Then, an extractor $\mathbf{E}$ can obtain an accepting transcript and reduced witnesses with expected running time $1/\varepsilon$ by providing a random challenge $x^* \xleftarrow{\$} \mathbb{F}^\times$ to $\mathbf{P}^*$. Let $\mathbf{w}_i^* = (\hat{\mathbf{F}}_i^*, \mathbf{R}_i, \hat{\mathbf{t}}_i, \mathbf{c}_i, \alpha_i)$ be the output witness from $\mathbf{P}^*$ for $0 \leq i < m_0$, and let $(y_g, x^*, \hat{y}_0^*, \dots, \hat{y}_{m_0-1}^*)$ be the corresponding transcript. Let

$$f_i^* = f_i + X^* \cdot g_i = \text{DcdToPoly}(\hat{\mathbf{F}}_i^*, \mathbf{c}_i, \alpha_i)$$

for $0 \leq i < m_0$, and define $f := \sum_{i=0}^{m_0-1} f_i \cdot V_{0,i}$ and $g := \sum_{i=0}^{m_0-1} g_i \cdot V_{0,i}$. The extractor then reconstructs the witness as $\mathbf{w} = (f, (\hat{\mathbf{F}}_i^*, \mathbf{R}_i, \hat{\mathbf{t}}_i, \mathbf{c}_i, \alpha_i)_{i=0}^{m_0-1})$. To ensure $(x, \mathbf{w}) \in R_{\text{Poly}}^{(B', B_o', B_c', B_\alpha')}$, it remains to show that $f(\mathbf{x}) = y$. By the verifier's check in the splitting protocol, we have

$$y + x^* \cdot y_g = \sum_{i=0}^{m_0-1} \langle \text{Dcd}(\hat{y}_i^*), \mathbf{w}_1 \rangle \cdot v_{0,i}.$$

Moreover, since $(x_i^*, \mathbf{w}_i^*) \in R_{\text{Quad}}^{(B', B_o', B_c', B_\alpha')}$, it follows that

$$f_i^*(x^*) = \langle \text{Dcd}(\hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}_i^* ((\alpha_i \mathbf{c}_i)^{-1} \circ \hat{\mathbf{w}}_0)), \mathbf{w}_1 \rangle = \langle \text{Dcd}(\hat{y}_i^*), \mathbf{w}_1 \rangle$$

for $0 \leq i < m_0$. Therefore,

$$f(\mathbf{x}) + x^* \cdot g(\mathbf{x}) = \sum_{i=0}^{m_0-1} f_i^*(x^*) \cdot v_{0,i} = \sum_{i=0}^{m_0-1} \langle \text{Dcd}(\hat{y}_i^*), \mathbf{w}_1 \rangle \cdot v_{0,i} = y + x^* \cdot y_g.$$

Equivalently,

$$f(\mathbf{x}) - y = x^* \cdot (y_g - g(\mathbf{x})).$$

If $f(\mathbf{x}) \neq y$, then this equation holds for at most one value of $x^* \in \mathbb{F}^\times$. Thus, by the Schwartz Zippel lemma, the probability that the verifier accepts while $f(\mathbf{x}) \neq y$ is at most $1/|\mathbb{F}^\times|$. Since $\varepsilon > 1/|\mathbb{F}^\times|$, the extractor obtains a valid witness with expected running time $1/\varepsilon = \text{poly}(\lambda)$. $\square$

B.8. Proof of Theorem 5

Proof. Public reducibility follows directly from the protocol. Below, we prove completeness and knowledge soundness.

Completeness. Consider $\begin{bmatrix} \sum_{i=0}^{m_0-1} \alpha_i \cdot \hat{\mathbf{F}}_i^* \\ \sum_{i=0}^{m_0-1} \alpha_i \cdot \mathbf{R}_i \end{bmatrix}$ as a vector over R . Its L^2 -norm is bounded by $m_0 \sqrt{n_0} B_c B$. Thus, by Lemma 2, the success probability in Line 14 is at least $(1 - 2^{-128})/M$.

For each $0 \leq k < n_0$, we have

$$\hat{\mathbf{f}}_k^* = \sum_{i=0}^{m_0-1} \alpha_i \hat{\mathbf{f}}_{i,k}^* + \hat{\mathbf{f}}_{m_0,k}^*, \quad \mathbf{r}_k = \sum_{i=0}^{m_0-1} \alpha_i \mathbf{r}_{i,k} + \mathbf{r}_{m_0,k},$$

$$\mathbf{h}_k = \sum_{i=0}^{m_0-1} \alpha_i \mathbf{h}_{i,k} + \mathbf{h}_{m_0,k}.$$

Therefore,

$$\begin{aligned} & \|\hat{\mathbf{f}}_k^*\|_2 + \|\mathbf{r}_k\|_2 + \|\mathbf{h}_k\|_2 \\ & \leq m_0 B_C B + \sigma(\sqrt{2(m_1+1)d} + \sqrt{2(\mu+\nu)d}) + B_t \sqrt{\mu d} = \underline{B}. \end{aligned}$$

Here,

$$\|\hat{\mathbf{f}}_{m_0,k}^*\|_2 + \|\mathbf{r}_{m_0,k}\|_2 \leq \sigma(\sqrt{2(m_1+1)d} + \sqrt{2(\mu+\nu)d})$$

with overwhelming probability by Lemma 1, and $\|\mathbf{h}_{m_0,k}\|_2 \leq B_t \sqrt{\mu d}$.

Similarly,

$$\|\mathbf{s}_2\|_2 + \|\hat{\mathbf{t}}\|_2 \leq m_0 B_C B_o + B_u \sqrt{\kappa d} + (q/B_t) \sqrt{\mu n_0 d} = \underline{B}_o.$$

This follows from

$$\hat{\mathbf{t}} = \sum_{i=0}^{m_0-1} \alpha_i \cdot \hat{\mathbf{t}}_i + \hat{\mathbf{t}}_{m_0}, \quad \mathbf{s} = \sum_{i=0}^{m_0-1} \alpha_i \cdot \mathbf{s}_i + \mathbf{s}_{m_0},$$

together with $\|\hat{\mathbf{t}}_{m_0}\|_2 \leq (q/B_t) \sqrt{\mu n_0 d}$ and $\|\mathbf{s}_{m_0}\|_2 \leq B_u \sqrt{\kappa d}$. Therefore, the input relation is reduced to $R_{\text{Quad}}^{(\underline{B}, \underline{B}_o, 1, 1)}$ with success probability at least $(1 - 2^{-128})/M$.

Knowledge Soundness. Suppose there exists a deterministic algorithm P^* that succeeds with non-negligible probability $\varepsilon > 2d\eta$. To construct an extractor for P^* , we define an algorithm E_i for $0 \leq i < m_0$ as follows.

$E_i^{P^*}(\text{pp}, x) :$

- 1) Run P^* with a random challenge $\alpha^{(i)} \xleftarrow{\$} \mathcal{C}^{m_0}$ until it succeeds, and obtain the accepting transcript $(\hat{\mathbf{u}}_{m_0}, \hat{y}_{m_0}, \alpha^{(i)})$ and reduced witness $(f^*, \hat{\mathbf{F}}^{*(i)}, \mathbf{R}^{(i)}, \hat{\mathbf{t}}^{(i)}, \mathbf{1}, 1)$. This takes expected time $1/\varepsilon$.
- 2) Run P^* at most $1/(\varepsilon/2 - d\eta)$ times with a challenge $\alpha'^{(i)}$ obtained from $\alpha^{(i)}$ by freshly sampling only the i -th coordinate from $\mathcal{C}$.
- 3) If no accepting transcript $(\hat{\mathbf{u}}_{m_0}, \hat{y}_{m_0}, \alpha'^{(i)})$ is obtained such that $\alpha_i^{(i)} - \alpha'_i \in R_p^\times$, abort.

Figure 9: Extractor for Π_{Agg}

Since ε is non-negligible in λ , the expected running time $1/\varepsilon + 1/(\varepsilon/2 - d\eta)$ of E_i is $\text{poly}(\lambda)$. To bound the abort probability, define a matrix $H_i \in \{0, 1\}^{|\mathcal{C}|^{m_0-1} \times |\mathcal{C}|}$ whose columns are indexed by the i -th component of α and whose rows are indexed by the remaining components. Each entry indicates whether P^* succeeds on the corresponding challenge. We say that a row of H_i is heavy if it contains more than a $\varepsilon/2$ fraction of ones. Then,

$$\Pr[\text{abort}] = \Pr[\text{abort} \mid \text{heavy}] \Pr[\text{heavy}] + \Pr[\text{abort} \mid \neg \text{heavy}] \Pr[\neg \text{heavy}].$$

By a simple counting argument, more than half of the accepting challenges lie in heavy rows, so $\Pr[\neg \text{heavy}] < 1/2$ when conditioning on the first accepting challenge. If α lies in a heavy row, then, by well spreadness, at least a $(\varepsilon/2 - d\eta)$ fraction of the freshly sampled i -th coordinates yield an accepting challenge whose difference from $\alpha_i^{(i)}$ is invertible. Thus,

$$\Pr[\text{abort} \mid \text{heavy}] \leq (1 - (\varepsilon/2 - d\eta))^{1/(\varepsilon/2 - d\eta)} < 1/e.$$

Therefore, $\Pr[\text{abort}] < 1/2 + 1/e$, which gives a success probability greater than $1/2 - 1/e > 0$. Thus, running E_i until it succeeds for each $0 \leq i < m_0$ results in an extractor E with expected running time $\text{poly}(\lambda, N)$ that outputs m_0 pairs

$$(\hat{\mathbf{u}}_{m_0}, \hat{y}_{m_0}, \alpha^{(i)}, x^{*(i)}, w^{*(i)}) \quad \text{and} \quad (\hat{\mathbf{u}}_{m_0}, \hat{y}_{m_0}, \alpha'^{(i)}, x'^{(i)}, w'^{(i)})$$

of accepting transcripts and reduced instance-witness pairs in $R_{\text{Quad}}^{(\underline{B}', \underline{B}_o', \underline{B}_c', 1)}$ for $0 \leq i < m_0$, where $\alpha^{(i)}$ and $\alpha'^{(i)}$ differ only in the i -th coordinate and their difference lies in $R_p^\times$. We note that $R_p^\times \subseteq R_{X^\gamma-b}^\times$ since $R_p = \mathbb{Z}_p[X]/(X^d + 1)$, $R_{X^\gamma-b} = \mathbb{Z}_p[X]/(X^\gamma - b)$, and $X^\gamma - b \mid X^d + 1$.

Next, we show that the witness w_i^* can be reconstructed from these transcripts for each $0 \leq i < m_0$. We first parse $x^{*(i)}, w^{*(i)}, x'^{(i)}, w'^{(i)}$ as follows.

$$x^{*(i)} = (\hat{\mathbf{u}}^{(i)}, x^*, \hat{y}^{*(i)}), \quad w^{*(i)} = (f^{*(i)}, \hat{\mathbf{F}}^{*(i)}, \mathbf{R}^{(i)}, \hat{\mathbf{t}}^{(i)}, \mathbf{c}^{(i)}, 1)$$

$$x'^{(i)} = (\hat{\mathbf{u}}^{(i)}, x^*, \hat{y}'^{*(i)}), \quad w'^{(i)} = (f'^{(i)}, \hat{\mathbf{F}}'^{(i)}, \mathbf{R}'^{(i)}, \hat{\mathbf{t}}'^{(i)}, \mathbf{c}'^{(i)}, 1).$$

For $0 \leq k < n_0$, define

- $\mathbf{s}^{(i)} := D\hat{\mathbf{t}}^{(i)} - B_u \hat{\mathbf{u}}^{(i)} \pmod{q_0}$,
- $\mathbf{h}_{i,k}^{(i)} := A\hat{\mathbf{f}}_k^{*(i)} + B\mathbf{r}_k^{(i)} - B_t c_k^{(i)} \hat{\mathbf{t}}_k^{(i)} \pmod{q}$,
- $\mathbf{s}'^{(i)} := D\hat{\mathbf{t}}'^{(i)} - B_u \hat{\mathbf{u}}'^{(i)} \pmod{q_0}$,
- $\mathbf{h}_k'^{(i)} := A\hat{\mathbf{f}}_k'^{(i)} + B\mathbf{r}_k'^{(i)} - B_t c_k'^{(i)} \hat{\mathbf{t}}_k'^{(i)} \pmod{q}$.

Let $\alpha_i := \alpha_i^{(i)} - \alpha'_i \in R_p^\times$. We reconstruct $w_i^* = (f_i^*, \hat{\mathbf{F}}_i^*, \mathbf{R}_i, \hat{\mathbf{t}}_i, \mathbf{c}_i, \alpha_i)$ as follows.

- $\mathbf{c}_i := \mathbf{c}^{(i)} \circ \mathbf{c}'^{(i)}$.
- $\hat{\mathbf{t}}_i := \hat{\mathbf{t}}^{(i)} - \hat{\mathbf{t}}'^{(i)}$.
- $\hat{\mathbf{f}}_{i,k}^* := c_k'^{(i)} \cdot \hat{\mathbf{f}}_k^{*(i)} - c_k^{(i)} \cdot \hat{\mathbf{f}}_k'^{(i)}$ for $0 \leq k < n_0$.
- $\mathbf{r}_{i,k} := c_k'^{(i)} \cdot \mathbf{r}_k^{(i)} - c_k^{(i)} \cdot \mathbf{r}_k'^{(i)}$ for $0 \leq k < n_0$.
- $\hat{\mathbf{F}}_i^* := [\hat{\mathbf{f}}_{i,0}^* \mid \cdots \mid \hat{\mathbf{f}}_{i,n_0-1}^*]$ and $\mathbf{R}_i := [\mathbf{r}_{i,0} \mid \cdots \mid \mathbf{r}_{i,n_0-1}]$.
- $f_i^* := \text{DcdToPoly}(\hat{\mathbf{F}}_i^*, \mathbf{c}_i, \alpha_i)$.

We also define

- $\mathbf{s}_i := \mathbf{s}^{(i)} - \mathbf{s}'^{(i)}$.
- $\mathbf{h}_{i,k} := c_k'^{(i)} \cdot \mathbf{h}_k^{(i)} - c_k^{(i)} \cdot \mathbf{h}_k'^{(i)}$ for $0 \leq k < n_0$.

Then, the following identities and bounds hold for $0 \leq i < m_0$.

- $\mathbf{s}_i = D\hat{\mathbf{t}}_i - B_u \alpha_i \cdot \hat{\mathbf{u}}_i \pmod{q_0}$.
- $\mathbf{h}_{i,k} = A\hat{\mathbf{f}}_{i,k}^* + B\mathbf{r}_{i,k} - B_t c_{i,k} \hat{\mathbf{t}}_{i,k} \pmod{q}$ for $0 \leq k < n_0$.
- $\|\mathbf{s}_i\|_2 + \|\hat{\mathbf{t}}_i\|_2 \leq 2\underline{B}_o' = \underline{B}_o'$.
- $\|\mathbf{c}_{i,k}\|_1 = \|c_k^{(i)} \cdot c_k'^{(i)}\|_1 \leq (\underline{B}_c')^2 = \underline{B}_c'$ for $0 \leq k < n_0$.
- $\|\alpha_i\|_1 = \|\alpha_i^{(i)} - \alpha_i'^{(i)}\|_1 \leq 2\underline{B}_c = \underline{B}_c'$.
- $\|\hat{\mathbf{f}}_{i,k}^*\|_2 + \|\mathbf{r}_{i,k}\|_2 + \|\mathbf{h}_{i,k}\|_2 \leq 2\underline{B}_c' \underline{B}' = \underline{B}'$ for $0 \leq k < n_0$.

Finally, it remains to show that

$$\hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}_i^* ((\alpha_i \mathbf{c}_i)^{-1} \circ \hat{\mathbf{w}}_0) = \hat{y}_i^* \pmod{X^\gamma - b}.$$

By definition, we have

$$\begin{aligned} \hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}_i^* (\mathbf{c}_i^{-1} \circ \hat{\mathbf{w}}_0) &= \hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}^{*(i)} ((\mathbf{c}^{(i)})^{-1} \circ \hat{\mathbf{w}}_0) - \hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}'^{(i)} ((\mathbf{c}'^{(i)})^{-1} \circ \hat{\mathbf{w}}_0) \\ &= \hat{y}^{*(i)} - \hat{y}'^{*(i)} = \alpha_i \cdot \hat{y}_i^* \pmod{X^\gamma - b}. \end{aligned}$$

Since $\alpha_i \in R_p^\times \subseteq R_{X^\gamma-b}^\times$, this implies

$$\hat{\mathbf{v}}_1^{*\top} \hat{\mathbf{F}}_i^* ((\alpha_i \mathbf{c}_i)^{-1} \circ \hat{\mathbf{w}}_0) = \hat{y}_i^* \pmod{X^\gamma - b}.$$

Therefore, $(x_i^*, w_i^*) \in R_{\text{Quad}}^{(\underline{B}', \underline{B}_o', \underline{B}_c', \underline{B}_\alpha')}$ for every $0 \leq i < m_0$. $\square$

B.9. Proof of Theorem 6

Proof. Below, we prove completeness and knowledge soundness of (Quad.P, Quad.V).

Completeness. Since $\bar{\mathbf{f}}^* = \sum_{k=0}^{n_0-1} c_k \hat{\mathbf{f}}_k^*$, $\bar{\mathbf{r}} = \sum_{k=0}^{n_0-1} c_k \mathbf{r}_k$, and $\bar{\mathbf{h}} = \sum_{k=0}^{n_0-1} c_k \mathbf{h}_k$, it holds that

$$\|\bar{\mathbf{f}}^*\|_2 + \|\bar{\mathbf{r}}\|_2 + \|\bar{\mathbf{h}}\|_2 \leq n_0 B_C B = \bar{B}.$$

Also, $\|\mathbf{s}\|_2 + \|\hat{\mathbf{t}}\|_2 \leq B_o = \bar{B}_o$ holds. Therefore, the protocol always accepts for an honest prover.

Knowledge Soundness. To prove knowledge soundness, we follow the approach of [BBC⁺18] with minor modifications. Suppose there exists a deterministic algorithm P^* that accepts with non-negligible probability $\varepsilon > 2d\eta$ for some instance $x^* = (\hat{\mathbf{u}}, x^*, \hat{y}^*)$. We first show that, for each $0 \leq k < n_0$, there exists an extractor E_k with expected running time $\text{poly}(\lambda)$ executions of P^* , which outputs two accepting transcripts $(\hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}^{(k)}, \bar{\mathbf{f}}^{*(k)}, \bar{\mathbf{r}}^{(k)})$ and $(\hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}'^{(k)}, \bar{\mathbf{f}}'^{(k)}, \bar{\mathbf{r}}'^{(k)})$ for the instance x^* with probability at least $\frac{1}{2} - \frac{1}{e}$, where $\mathbf{c}^{(k)}$ and $\mathbf{c}'^{(k)}$ differ only in the k -th coordinate and $c_k^{(k)} - c_k'^{(k)}$ is invertible. We construct the extractor E_k as follows.

$E_k^{P^*}(\text{pp}, x^*) :$

- 1) Run P^* with a random challenge $\mathbf{c}^{(k)} \xleftarrow{\$} \mathcal{C}^{n_0}$ until it succeeds, and obtain the accepting transcript $(\hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}^{(k)}, \bar{\mathbf{f}}^{*(k)}, \bar{\mathbf{r}}^{(k)})$. This takes expected time $1/\varepsilon$.
- 2) Run P^* at most $1/(\varepsilon/2 - d\eta)$ times with a challenge $\mathbf{c}'^{(k)}$ obtained from $\mathbf{c}^{(k)}$ by freshly sampling only the k -th coordinate from $\mathcal{C}$.
- 3) If no accepting transcript $(\hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}'^{(k)}, \bar{\mathbf{f}}'^{(k)}, \bar{\mathbf{r}}'^{(k)})$ is obtained such that $c_k^{(k)} - c_k'^{(k)} \in R_p^\times$, abort.

Figure 10: Extractor for Π_{Quad}

Since ε is non-negligible in λ , the expected running time $1/\varepsilon + 1/(\varepsilon/2 - d\eta)$ of E_k is $\text{poly}(\lambda)$. To bound the abort probability, define a matrix $H_k \in \{0, 1\}^{|\mathcal{C}|^{n_0-1} \times |\mathcal{C}|}$ whose columns are indexed by the k -th component of $\mathbf{c}$ and whose rows are indexed by the remaining components. Each entry indicates whether P^* succeeds on the corresponding challenge. We say that a row of H_k is heavy if it contains more than a $\varepsilon/2$ fraction of ones. Then,

$$\Pr[\text{abort}] = \Pr[\text{abort} \mid \text{heavy}] \Pr[\text{heavy}] + \Pr[\text{abort} \mid \neg \text{heavy}] \Pr[\neg \text{heavy}].$$

By a simple counting argument, more than half of the accepting challenges lie in heavy rows, so $\Pr[\neg \text{heavy}] < 1/2$ when conditioning on the first accepting challenge. If $\mathbf{c}$ lies in a heavy row, then, by well spreadness, at least a $(\varepsilon/2 - d\eta)$ fraction of the freshly sampled k -th coordinates yield an accepting challenge whose difference from $c_k^{(k)}$ is invertible. Thus,

$$\Pr[\text{abort} \mid \text{heavy}] \leq (1 - (\varepsilon/2 - d\eta))^{1/(\varepsilon/2 - d\eta)} < 1/e.$$

Therefore, $\Pr[\text{abort}] < 1/2 + 1/e$, which gives a success probability greater than $1/2 - 1/e > 0$. Thus, running E_k until it succeeds for each $0 \leq k < n_0$ results in an

extractor E with expected running time $\text{poly}(\lambda, N)$ that outputs n_0 pairs of accepting transcripts $(\hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}^{(k)}, \bar{\mathbf{f}}^{*(k)}, \bar{\mathbf{r}}^{(k)})$ and $(\hat{\mathbf{t}}, \mathbf{z}, \mathbf{c}'^{(k)}, \bar{\mathbf{f}}'^{(k)}, \bar{\mathbf{r}}'^{(k)})$ for $0 \leq k < n_0$, where $\mathbf{c}^{(k)}$ and $\mathbf{c}'^{(k)}$ differ only in the k -th coordinate and their difference lies in $R_p^\times$. We note that $R_p^\times \subseteq R_{X^\gamma-b}^\times$ since $R_p = \mathbb{Z}_p[X]/(X^d + 1)$, $R_{X^\gamma-b} = \mathbb{Z}_p[X]/(X^\gamma - b)$, and $X^\gamma - b \mid X^d + 1$.

We now show that a witness $w^* = (f^*, \text{op}^*)$ can be reconstructed from these transcripts. Let $\mathbf{s} := \mathbf{D}\hat{\mathbf{t}} - B_u \hat{\mathbf{u}}$. Then, by definition,

$$\|\mathbf{s}\|_2 + \|\hat{\mathbf{t}}\|_2 \leq \bar{B}_o = B'_o.$$

Next, let

$$\bar{\mathbf{h}}^{(k)} := \mathbf{A}\bar{\mathbf{f}}^{*(k)} + \mathbf{B}\bar{\mathbf{r}}^{(k)} - B_t \cdot \hat{\mathbf{T}}\mathbf{c}^{(k)}$$

and

$$\bar{\mathbf{h}}'^{(k)} := \mathbf{A}\bar{\mathbf{f}}'^{(k)} + \mathbf{B}\bar{\mathbf{r}}'^{(k)} - B_t \cdot \hat{\mathbf{T}}\mathbf{c}'^{(k)}$$

for $0 \leq k < n_0$. Define c_k , $\hat{\mathbf{f}}_k^*$, $\mathbf{r}_k$, and $\mathbf{h}_k$ as follows.

$$\begin{aligned} c_k &:= c_k^{(k)} - c_k'^{(k)}, & \hat{\mathbf{f}}_k^* &:= \bar{\mathbf{f}}^{*(k)} - \bar{\mathbf{f}}'^{(k)}, \\ \mathbf{r}_k &:= \bar{\mathbf{r}}^{(k)} - \bar{\mathbf{r}}'^{(k)}, & \mathbf{h}_k &:= \bar{\mathbf{h}}^{(k)} - \bar{\mathbf{h}}'^{(k)}. \end{aligned}$$

Then, for $0 \leq k < n_0$, we have

$$\mathbf{h}_k = \mathbf{A}\hat{\mathbf{f}}_k^* + \mathbf{B}\mathbf{r}_k - B_t c_k \cdot \hat{\mathbf{t}}_k,$$

and

$$\|c_k\|_1 \leq 2B_C = B'_C, \quad \|\hat{\mathbf{f}}_k^*\|_2 + \|\mathbf{r}_k\|_2 + \|\mathbf{h}_k\|_2 \leq 2\bar{B} = B'.$$

Thus, $\text{Open}^{*(B, B_o, B_c, B_\alpha)}(\text{pp}, \hat{\mathbf{u}}, f^*, \text{op}^*) = 1$ for $\text{op}^* = (\hat{\mathbf{F}}^*, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{c}, 1)$ and $f^* := \text{DcdToPoly}(\hat{\mathbf{F}}^*, \mathbf{c}, 1)$.

It remains to show that $\hat{\mathbf{v}}_1^\top \hat{\mathbf{F}}^*(\mathbf{c}^{-1} \circ \hat{\mathbf{w}}_0) = \hat{y}^*$. Let $\mathbf{z} = (z_0, \dots, z_{n_0-1})$. For $0 \leq k < n_0$, the two accepting transcripts satisfy

$$\hat{\mathbf{v}}_1^\top \bar{\mathbf{f}}^{*(k)} = \mathbf{z}^\top \mathbf{c}^{(k)} \quad \text{and} \quad \hat{\mathbf{v}}_1^\top \bar{\mathbf{f}}'^{(k)} = \mathbf{z}^\top \mathbf{c}'^{(k)}.$$

Since $\mathbf{c}^{(k)}$ and $\mathbf{c}'^{(k)}$ differ only in the k -th coordinate, subtracting the two equations gives $\hat{\mathbf{v}}_1^\top \hat{\mathbf{f}}_k^* = c_k z_k$. Then, we have

$$\hat{\mathbf{v}}_1^\top \hat{\mathbf{F}}^*(\mathbf{c}^{-1} \circ \hat{\mathbf{w}}_0) = (\mathbf{c} \circ \mathbf{z})^\top (\mathbf{c}^{-1} \circ \hat{\mathbf{w}}_0) = \mathbf{z}^\top \hat{\mathbf{w}}_0 = \hat{y}^* \pmod{X^\gamma - b}.$$

Therefore, $((\hat{\mathbf{u}}, x^*, \hat{y}^*), (f^*, \text{op}^*)) \in R_{\text{Quad}}^{(B', B'_o, B'_c, 1)}$ holds, where $B' = 2\bar{B}$, $B'_o = \bar{B}_o$, and $B'_c = 2B_C$. $\square$

B.10. Proof of Theorem 7

Proof. To prove evaluation hiding, we construct a simulator $S = (S_0, S_1)$ as follows.

- $S_0(\text{pp}) \rightarrow (\text{com} = (\hat{\mathbf{u}}_i)_{i=0}^{m_0-1}, \text{st} = (\mathbf{T}_i, \hat{\mathbf{t}}_i)_{i=0}^{m_0-1}) :$

- 1) Sample $\mathbf{T}_i = [\mathbf{t}_{i,0} \mid \dots \mid \mathbf{t}_{i,n_0-1}] \xleftarrow{\$} R_q^{\mu \times n_0}$, and compute $\hat{\mathbf{t}}_i = \lfloor \mathbf{t}_{i,0}/B_t \rfloor \parallel \dots \parallel \lfloor \mathbf{t}_{i,n_0-1}/B_t \rfloor \in R^{\mu n_0}$ for $0 \leq i < m_0$.
- 2) Compute $\hat{\mathbf{u}}_i = \lfloor \mathbf{D}\hat{\mathbf{t}}_i/B_u \rfloor \in R^\kappa$ for $0 \leq i < m_0$.

$$\begin{aligned}
& \bullet S_1 \left(\text{pp}, \text{st} = (\mathbf{T}_i, \hat{\mathbf{t}}_i)_{i=0}^{m_0-1}, \mathbf{x} \right) \\
& \rightarrow (y_g, x^*, (\hat{y}_i^*)_{i=0}^{m_0-1}, \hat{\mathbf{u}}_{m_0}, \hat{y}_{m_0}, \boldsymbol{\alpha}, f^*, \hat{\mathbf{F}}^*, \mathbf{R}, \hat{\mathbf{t}}) : \\
& 1) \text{ Sample } x^* \xleftarrow{\$} \mathbb{F}^\times \text{ and } \hat{y}_0^*, \dots, \hat{y}_{m_0-1}^* \xleftarrow{\$} R_{X^\gamma-b}, \text{ and set} \\
& \quad y_g := (x^*)^{-1} \cdot \left(\sum_{i=0}^{m_0-1} \langle \text{Dcd}(\hat{y}_i^*), \mathbf{w}_1 \rangle \cdot v_{0,i} - y \right). \\
& 2) \text{ Sample } \boldsymbol{\alpha} \xleftarrow{\$} \mathcal{C}^{m_0} \text{ and} \\
& \quad \begin{bmatrix} \hat{\mathbf{F}}^* \\ \mathbf{R} \end{bmatrix} \xleftarrow{\$} \mathcal{D}_\sigma^{(m_1+\mu+\nu+1) \times n_0 d}. \\
& 3) \text{ Compute} \\
& \quad \mathbf{T}_{m_0} := [\mathbf{t}_{m_0,0} \mid \dots \mid \mathbf{t}_{m_0,n_0-1}] \\
& \quad = \mathbf{A}\hat{\mathbf{F}}^* + \mathbf{B}\mathbf{R} - \sum_{i=0}^{m_0-1} \alpha_i \cdot \mathbf{T}_i \pmod{q}, \\
& \quad \text{and compute } \hat{\mathbf{t}}_{m_0} = \lfloor \mathbf{t}_{m_0,0}/B_t \rfloor \parallel \dots \parallel \lfloor \mathbf{t}_{m_0,n_0-1}/B_t \rfloor \in R^{\mu n_0}. \\
& 4) \text{ Compute } \hat{\mathbf{u}}_{m_0} = \lfloor \mathbf{D}\hat{\mathbf{t}}_{m_0}/B_u \rfloor \in R^\kappa \text{ and } \hat{\mathbf{t}} := \sum_{i=0}^{m_0-1} \alpha_i \cdot \hat{\mathbf{t}}_i + \hat{\mathbf{t}}_{m_0} \in R^{\mu n_0}. \\
& 5) \text{ Compute } f^* = \text{DcdToPoly}(\hat{\mathbf{F}}^*, \mathbf{1}, 1) \text{ and} \\
& \quad \hat{y}_{m_0}^* := \hat{\mathbf{v}}_1^\top \hat{\mathbf{F}}^* \hat{\mathbf{w}}_0 - \sum_{i=0}^{m_0-1} \alpha_i \cdot \hat{y}_i^* \pmod{X^\gamma-b}.
\end{aligned}$$

For S_0 , the simulator outputs $\text{st} = (\mathbf{T}_i, \hat{\mathbf{t}}_i)_{i=0}^{m_0-1}$ as the intermediate state passed to S_1 . For S_1 , it suffices to simulate the transcript $(y_g, x^*, (\hat{y}_i^*)_{i=0}^{m_0-1}, \hat{\mathbf{u}}_{m_0}, \hat{y}_{m_0}, \boldsymbol{\alpha})$ for $\Pi_{\text{Split}} \circ \Pi_{\text{Agg}}$ and the reduced witness $\mathbf{w}^* = (f^*, \hat{\mathbf{F}}^*, \mathbf{R}, \hat{\mathbf{t}}, \mathbf{1}, 1)$, since the transcript of Π_{Quad} can be simulated from these values.

Now, let A be a PPT adversary for the evaluation hiding property. We prove evaluation hiding using the following hybrid argument.

Hybrid Argument 0. In this hybrid argument, we prove the following.

$$\begin{aligned}
& \Pr \left[\begin{array}{l} \mathbf{A}(\text{pp}, \text{com}, \text{tr}) = 1 \wedge \\ b = 1 \end{array} \middle| \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, D) \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f) \\ \mathbf{x} \leftarrow \mathbf{A}(\text{pp}, \text{com}) \\ \mathbf{x} := (\text{com}, \mathbf{x}, f(\mathbf{x})), \mathbf{w} := (f, \text{op}) \\ (\text{tr}, b) \leftarrow (\text{Eval.P}(\text{pp}, \mathbf{x}, \mathbf{w}), \text{Eval.V}(\text{pp}, \mathbf{x})) \end{array} \right] \\
& - \Pr \left[\mathbf{A}(\text{pp}, \text{com}, \text{tr}) = 1 \middle| \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, D) \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f) \\ \mathbf{x} \leftarrow \mathbf{A}(\text{pp}, \text{com}) \\ \mathbf{x} := (\text{com}, \mathbf{x}, f(\mathbf{x})) \\ \text{st} := (\mathbf{T}_i, \hat{\mathbf{t}}_i)_{i=0}^{m_0-1} \\ \text{tr} \leftarrow S_1(\text{pp}, \text{st}, \mathbf{x}) \end{array} \right] \leq 2^{-128}.
\end{aligned}$$

Since com follows the same distribution in both experiments, it suffices to compare the distributions of the accepting transcripts tr . For an accepting transcript of Π_{Eval} , all remaining values are determined by $(x^*, y_g, (\hat{y}_i^*)_{i=0}^{m_0-1}, \boldsymbol{\alpha}, \hat{\mathbf{F}}^*, \mathbf{R})$ generated during the protocol. By construction, x^* is uniform over $\mathbb{F}^\times$, and $\boldsymbol{\alpha}$ is uniform over $\mathcal{C}^{m_0}$. Similarly, $(\hat{y}_i^*)_{i=0}^{m_0-1}$ is uniform over $R_{X^\gamma-b}^{m_0}$, since the masking vectors $\hat{\mathbf{g}}_i$ are sampled uniformly from $R_{X^\gamma-b}^{n_0}$ during the splitting step of Com , and the induced masks $\hat{\mathbf{g}}_i^\top \hat{\mathbf{w}}_0$ are uniform over $R_{X^\gamma-b}$ because $w_{0,0} = 1$. Once

$(\hat{y}_i^*)_{i=0}^{m_0-1}$ and x^* are fixed, the value y_g is uniquely determined by the verifier equation. Finally, conditioned on these values, the accepted distribution of $(\hat{\mathbf{F}}^*, \mathbf{R})$ is within statistical distance 2^{-128} of $\mathcal{D}_\sigma^{(m_1+\mu+\nu+1) \times n_0 d}$ by Lemma 2. Therefore, the distributions of $(x^*, y_g, (\hat{y}_i^*)_{i=0}^{m_0-1}, \boldsymbol{\alpha}, \hat{\mathbf{F}}^*, \mathbf{R})$ in the real protocol and in the simulation by S_1 are statistically close. As a result, the accepting transcript distributions in the real and simulated experiments are within statistical distance at most 2^{-128} .

Hybrid Argument 1. In this hybrid argument, we prove the following.

$$\begin{aligned}
& \Pr \left[\mathbf{A}(\text{pp}, \text{com}, \text{tr}) = 1 \middle| \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, D) \\ (\text{com}, \text{op}) \leftarrow \text{Com}(\text{pp}, f) \\ \mathbf{x} \leftarrow \mathbf{A}(\text{pp}, \text{com}) \\ \mathbf{x} := (\text{com}, \mathbf{x}, f(\mathbf{x})) \\ \text{st} := (\mathbf{T}_i, \hat{\mathbf{t}}_i)_{i=0}^{m_0-1} \\ \text{tr} \leftarrow S_1(\text{pp}, \text{st}, \mathbf{x}) \end{array} \right] \\
& - \Pr \left[\mathbf{A}(\text{pp}, \text{com}, \text{tr}) = 1 \middle| \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, D) \\ (\text{com}, \text{st}) \leftarrow S_0(\text{pp}) \\ \mathbf{x} \leftarrow \mathbf{A}(\text{pp}, \text{com}) \\ \mathbf{x} := (\text{com}, \mathbf{x}, f(\mathbf{x})) \\ \text{tr} \leftarrow S_1(\text{pp}, \text{st}, \mathbf{x}) \end{array} \right] = \text{negl}(\lambda).
\end{aligned}$$

Since tr is generated by S_1 using only $(\mathbf{x}, \text{st})$, it suffices to show that the distributions of $(\text{com}, \mathbf{x}, \text{st})$ are computationally indistinguishable in the two experiments. For $\text{com} = (\hat{\mathbf{u}}_i)_{i=0}^{m_0-1}$, each $\hat{\mathbf{u}}_i$ is determined by the rounded inner commitments $\hat{\mathbf{t}}_i$, which are in turn derived from $\mathbf{T}_i$. In the real experiment, $\mathbf{T}_i = \mathbf{A}\hat{\mathbf{F}}_i^* + \mathbf{B}\mathbf{R}_i \pmod{q}$ for $0 \leq i < m_0$. In the simulated experiment, $\mathbf{T}_i$ is sampled uniformly from $R_q^{\mu \times n_0}$. By the hardness of $\text{MLWE}_{q,\nu,\chi}$, the distributions of $(\mathbf{T}_i)_{i=0}^{m_0-1}$ in the real and simulated experiments are computationally indistinguishable. Therefore, the distributions of $(\hat{\mathbf{t}}_i)_{i=0}^{m_0-1}$ and $\text{com} = (\hat{\mathbf{u}}_i)_{i=0}^{m_0-1}$ are also computationally indistinguishable. Since the evaluation point $\mathbf{x}$ is chosen adaptively by A from (pp, com) , the joint distributions of $(\text{com}, \mathbf{x}, \text{st})$ are computationally indistinguishable in the two experiments.

Combining Hybrid Argument 0 and Hybrid Argument 1, the PPT adversary A can distinguish the real distribution of (com, tr) from the simulated distribution with advantage at most $\text{negl}(\lambda) + 2^{-128}$. This proves evaluation hiding. $\square$

B.11. Proof of Theorem 8

Proof. We analyze the computational complexity of the prover and verifier, and the communication cost of Π_{Eval} in Fig. 7.

Commitment. For the time complexity of the commitment algorithm Com , it suffices to analyze the time complexity of Com^* , as Com iterates Com^* for n_0 times. The time complexity of Com^* is dominated by the generation of $\hat{\mathbf{t}}$, which takes $O(\mu m n_1)$. Thus, the total time complexity is $O(\mu m n_0 n_1) = O(\mu N)$. We note that $O(\mu) = O(\log^2 N)$ following the security analysis in Eq. (2). Therefore, the total time complexity is $O(N \log^2 N)$.

Evaluation. The evaluation complexity is dominated by the generation of $\hat{\mathbf{F}}$ in Π_{Agg} , which takes $O(m n_0 n_1) = O(N)$.

Verifier. Π_{Split} takes $O(n_0)$ time to generate y_0^* , and Π_{Agg} takes $O(\kappa n_0)$ time to generate $\hat{\mathbf{u}}$. Finally, Π_{SubEval} takes $O(\kappa \mu n_1)$ to compute $\mathbf{s}$ and $O(\mu m)$ to compute $\mathbf{h}$. We note that $O(\kappa) = O(\mu) = O(\log^2 N)$ following the security analysis in Eq. (2). Therefore, the total time complexity of the verifier is $O(m \log^2 N + n_0 \log^2 N + n_1 \log^4 N)$.

Communication. The commitment requires $O(\kappa n_0)$ communication complexity. For Π_{SubEval} , $\hat{\mathbf{t}}$ requires $O(\mu n_1)$ and $\bar{\mathbf{f}}$ requires $O(m)$. We note that $O(\kappa) = O(\mu) = O(\log^2 N)$ following the security analysis in Eq. (2). Therefore, the total communication complexity is $O(m + n_0 \log^2 N + n_1 \log^2 N)$.

Optimum. The verification and communication complexities depend on the choices of m , n_0 and n_1 , which satisfy $mn_0n_1 = N/n_2$. When focusing on communication cost, the optimum is obtained when $m = O(N^{1/3} \cdot \log^{4/3} N)$, $n_0 = O(N^{1/3}/\log^{2/3} N)$ and $n_1 = O(N^{1/3}/\log^{2/3} N)$, which results in communication complexity $O(N^{1/3} \log^{4/3} N)$. $\square$

Appendix C. Batched Evaluation

We present the batched splitting protocol, which enables batched polynomial evaluation during PIOP compilation.

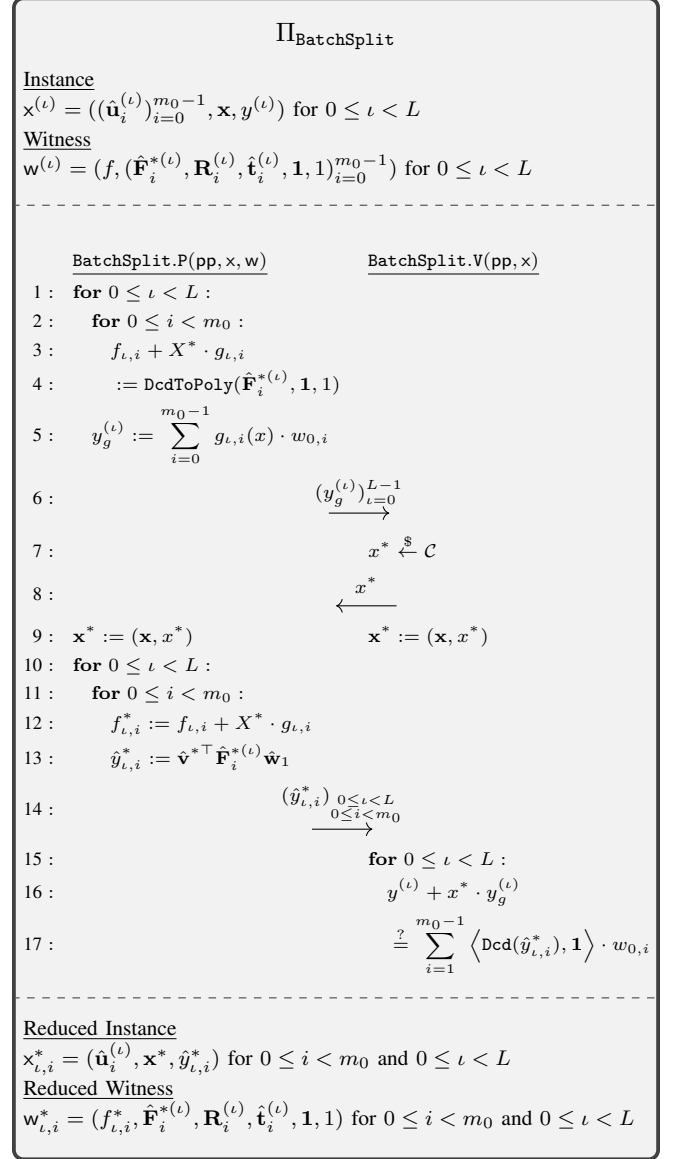


Figure 11: Batched Splitting Protocol of Jindo